Geocomputation

Accessibility Analysis



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Transport network and accessibility analysis.
- Spatial network structure.
- Dijkstra’s shortest path algorithm.
- An example of transport network and accessibility analysis.

Spatial interaction

- We have mostly dealt with static events: events that happen at a particular point in time and space.
- Spatial interaction is concerned with the idea that there are relations between different locations (e.g. measured in people or goods travelling).

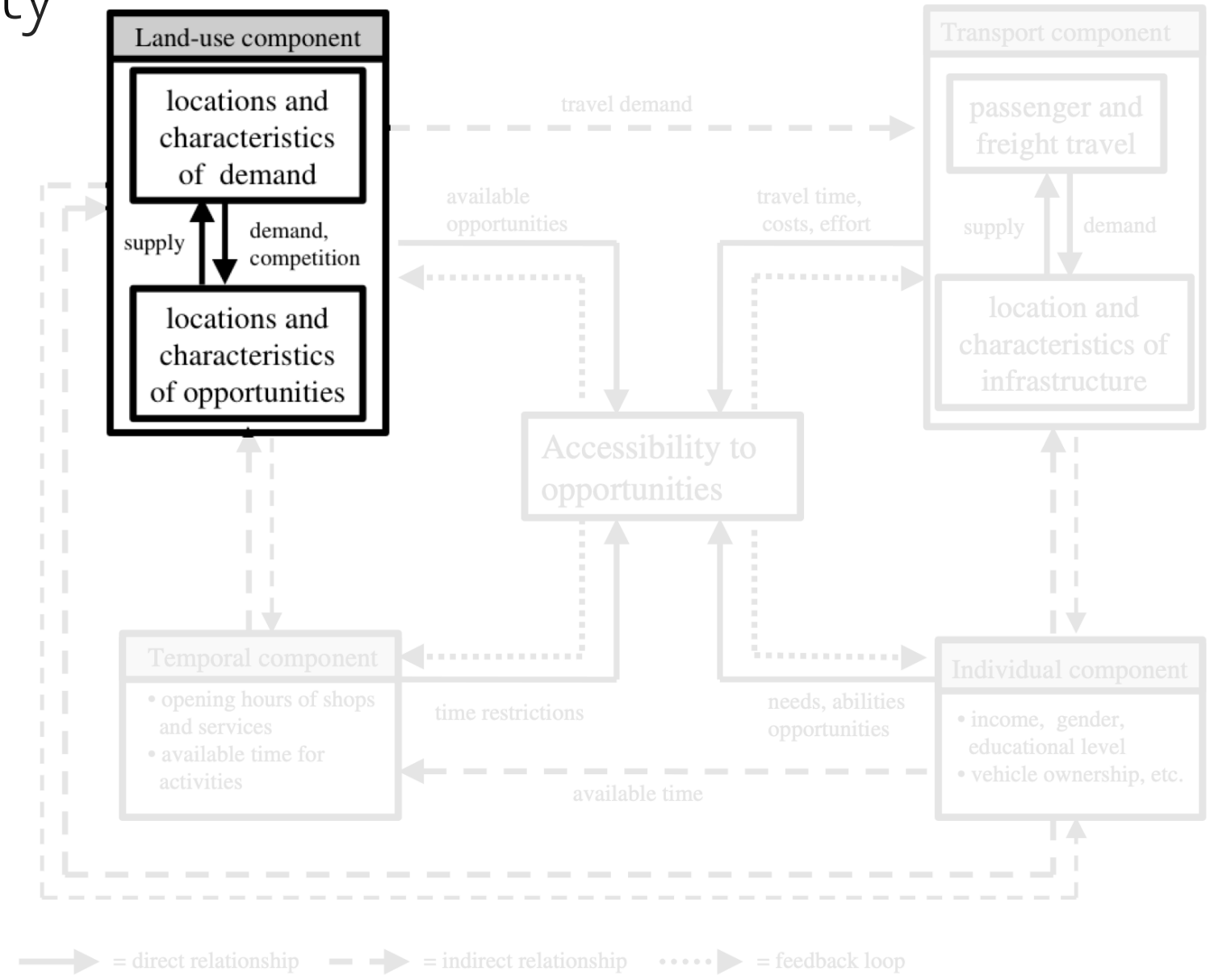
Transport links actors and activities



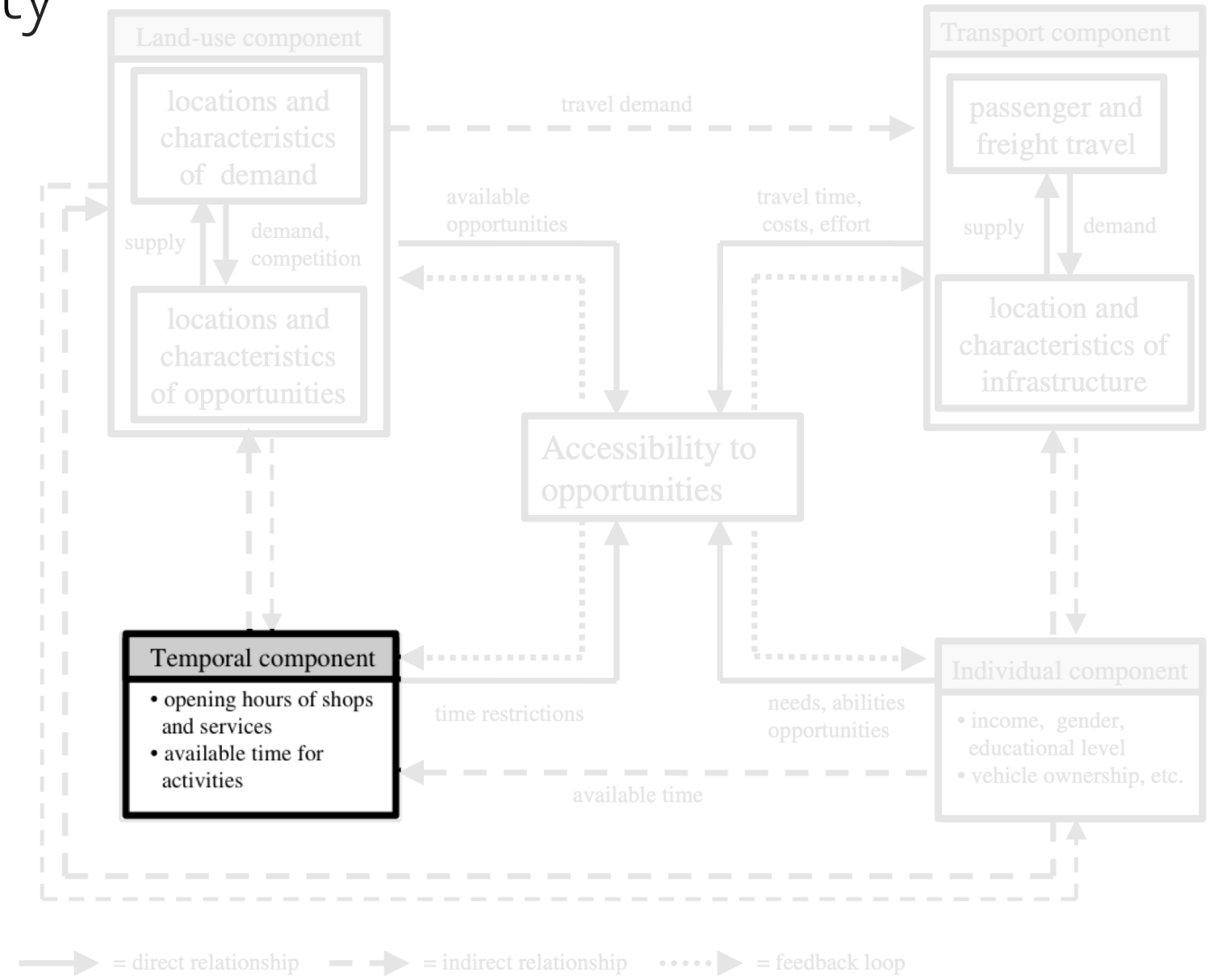
Time geography

- Proposed by Torsten Hägerstrand in the 1970s.
- Time geography describes the life of an individual as a continuous path through time and space, constituted by movements through space and activities localised in space.
- All activities are governed by three constraints: capability constraints, coupling constraints, and authority constraints.
- These constraints can mitigate but also reinforce one another's impacts on activity participation and travel behaviour.
- When people meet their individual space-time paths form a bundle.

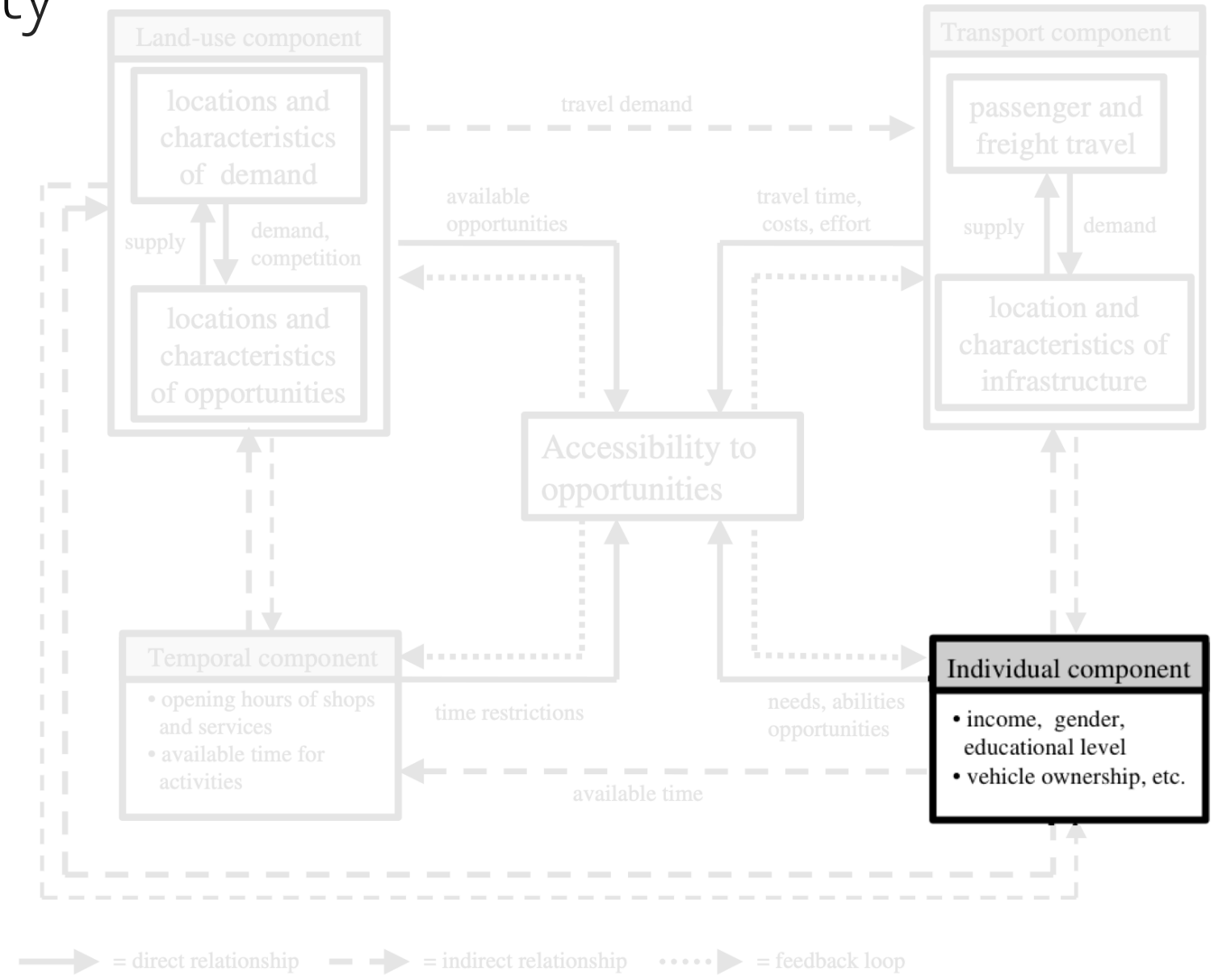
Accessibility



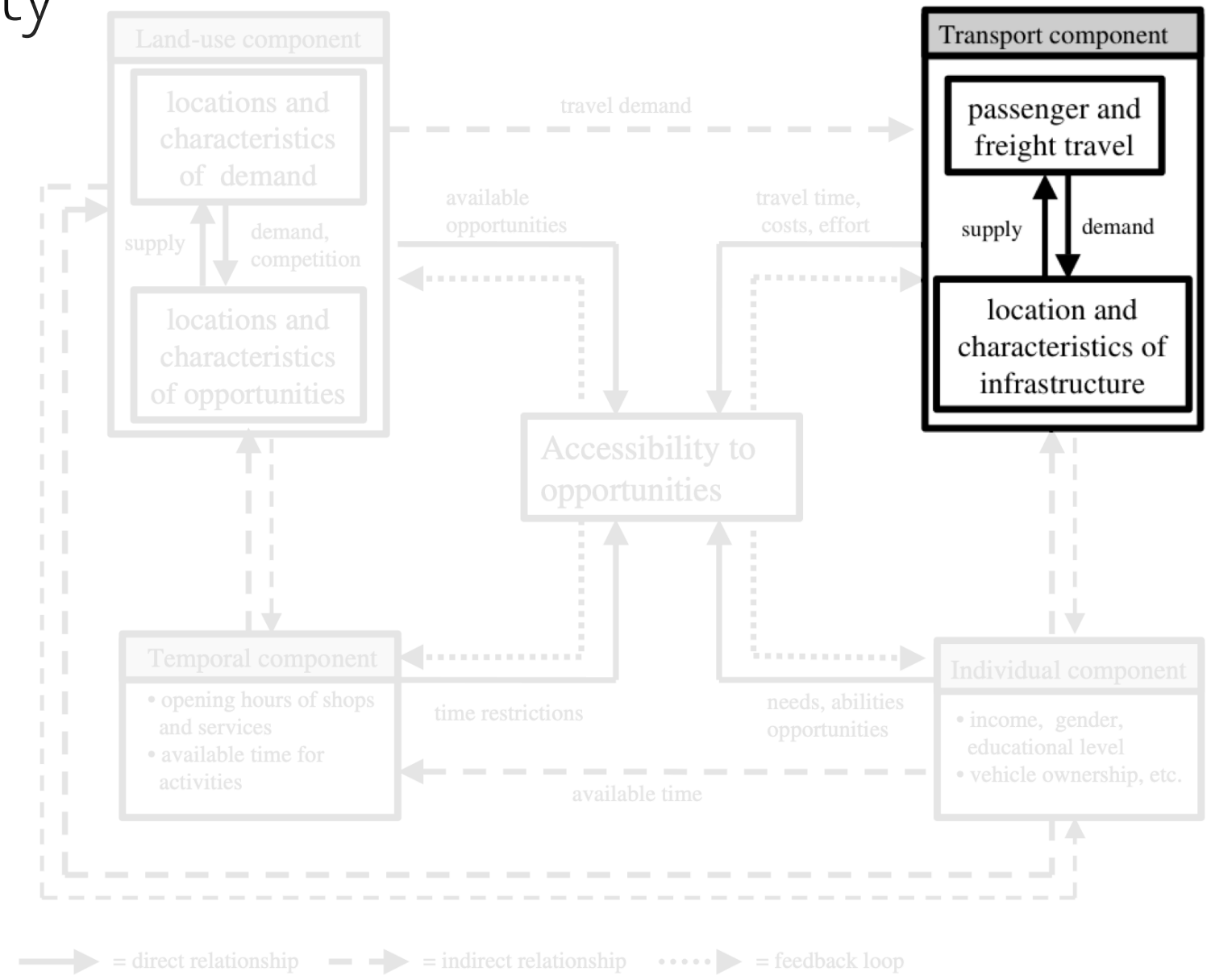
Accessibility



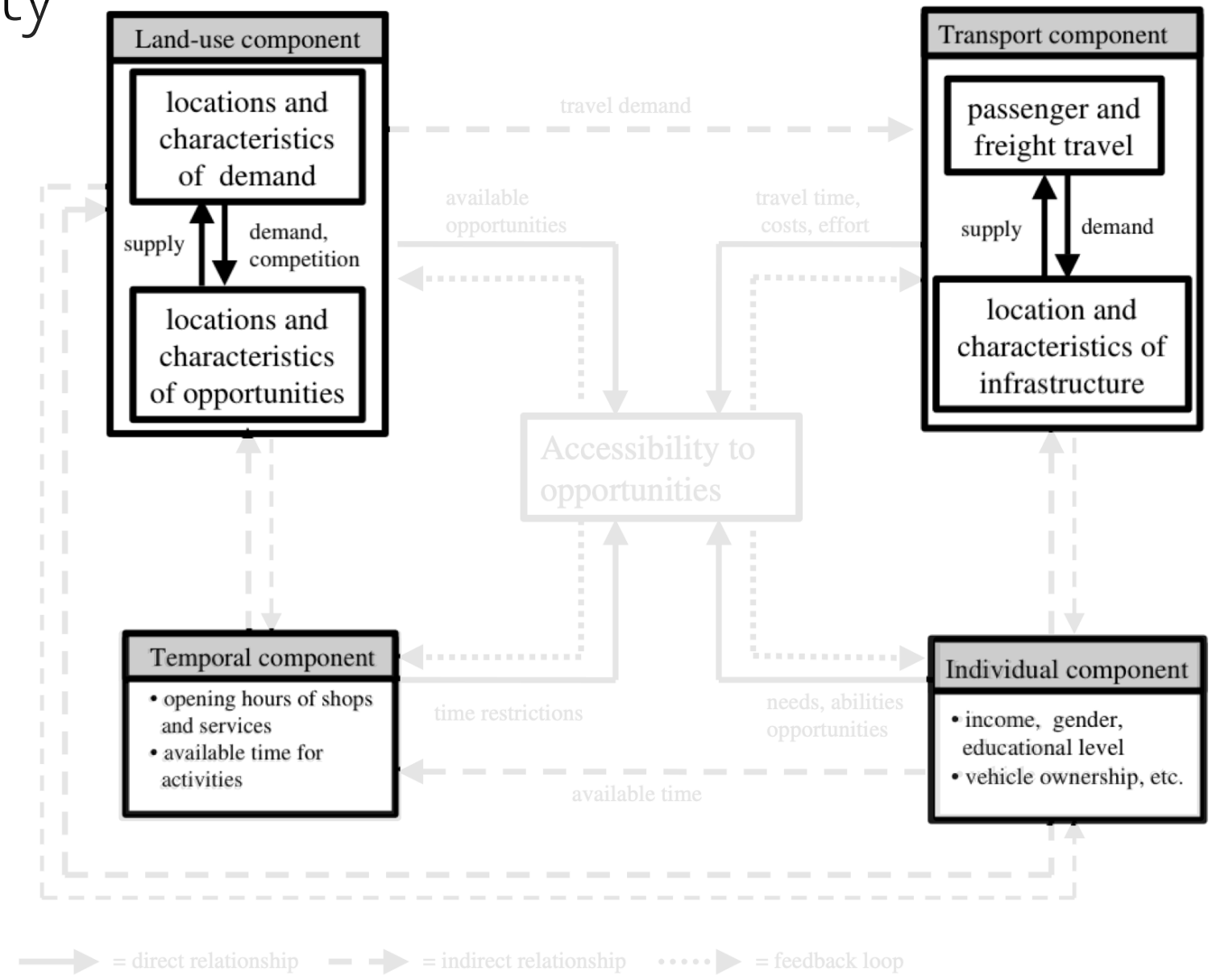
Accessibility



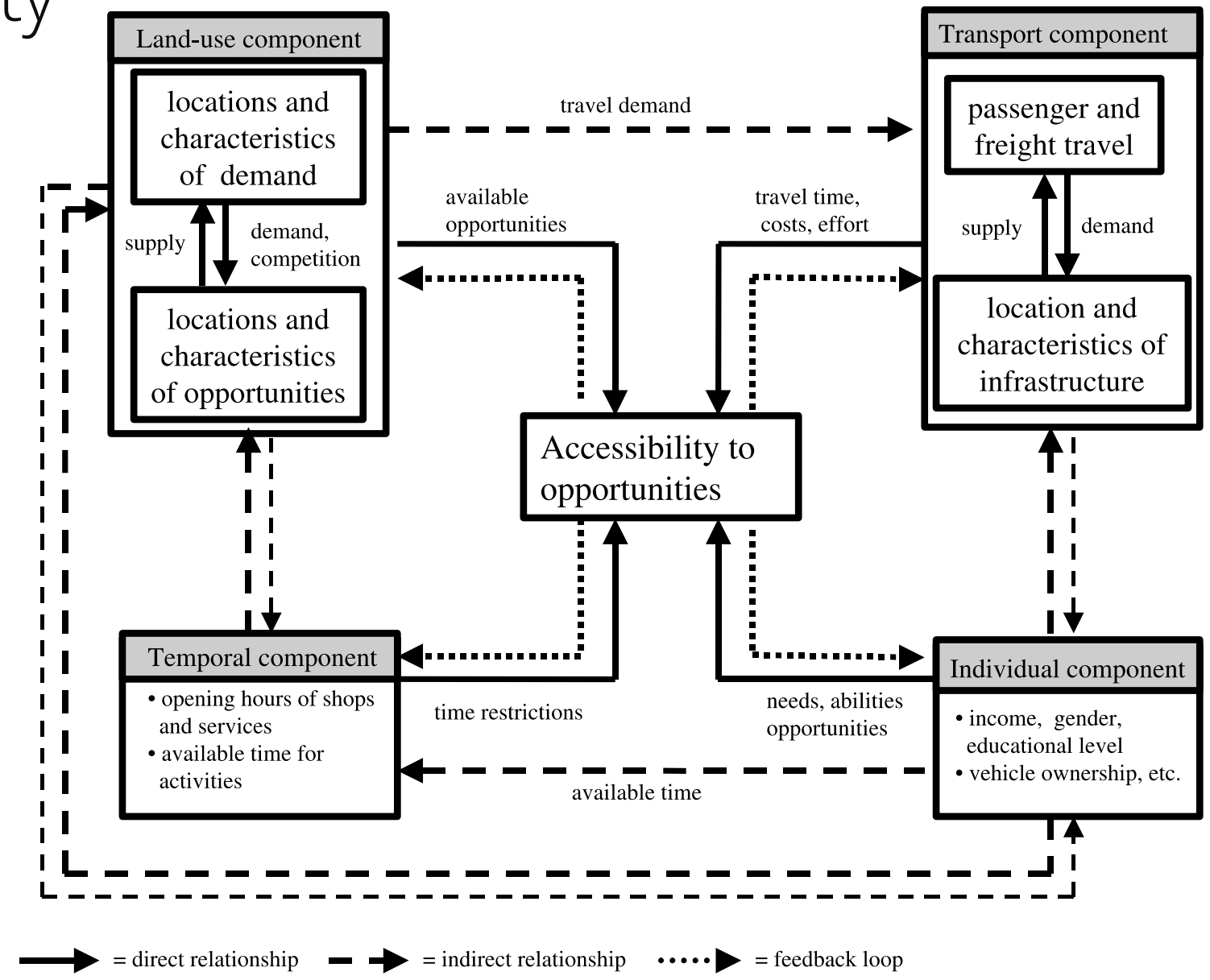
Accessibility



Accessibility

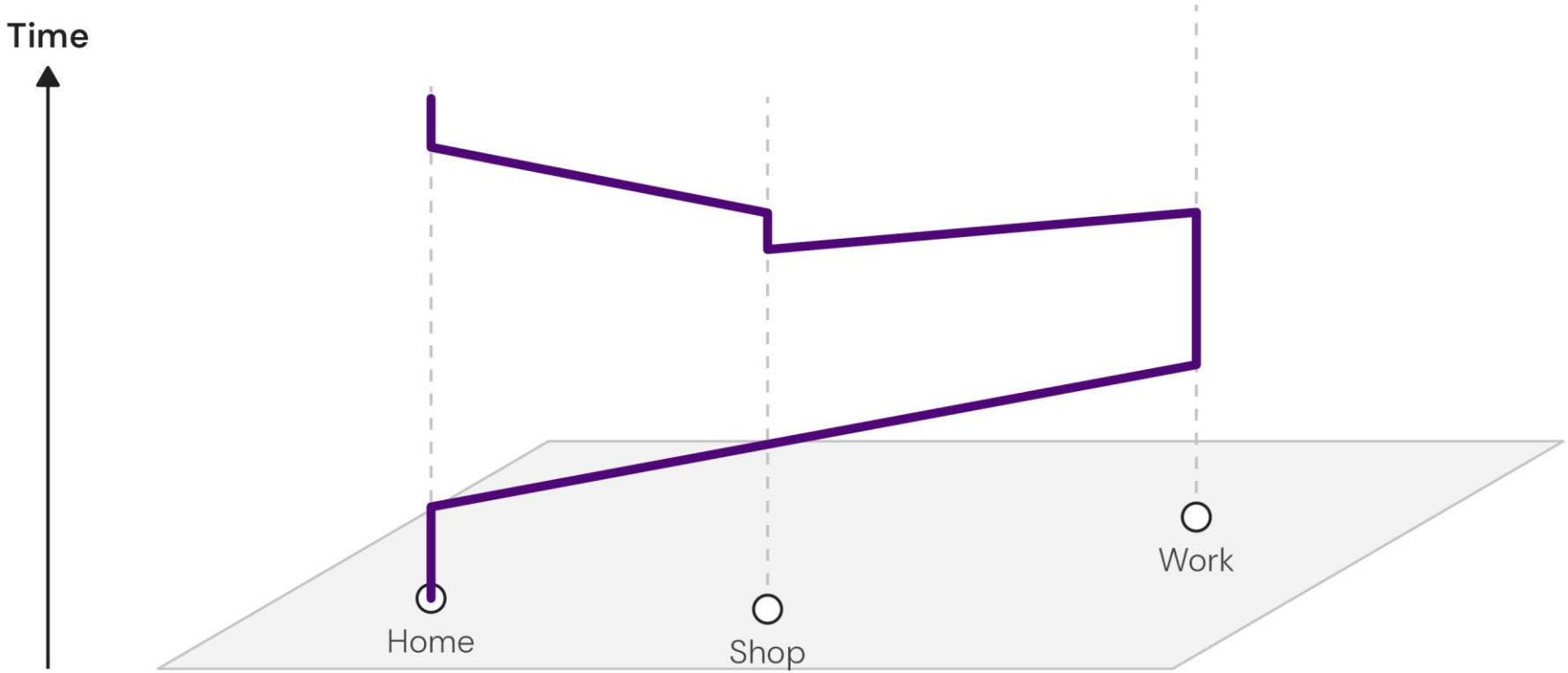


Accessibility

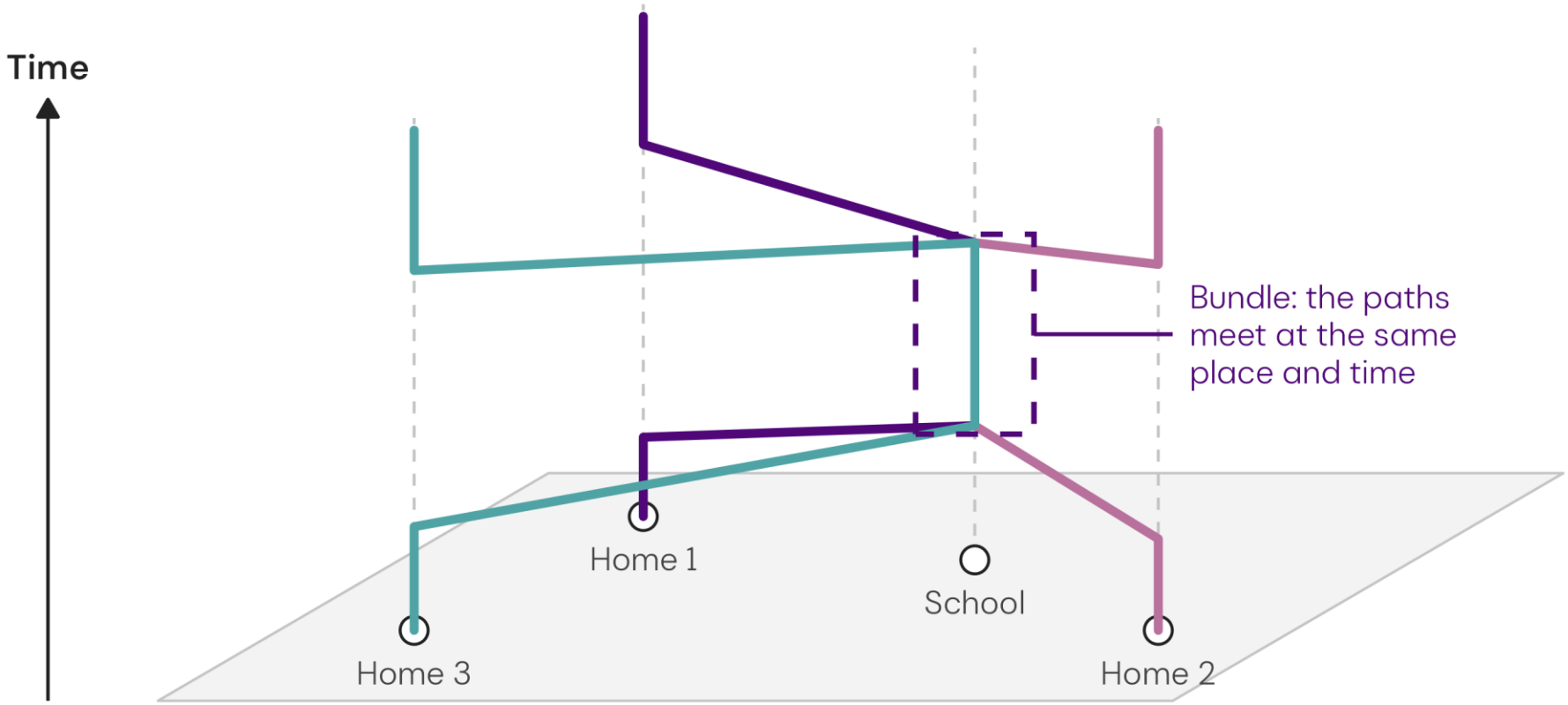


Geurs and Van Wee 2004

Time geography: a path through space and time



Bundles: when paths meet



Typical accessibility questions

Typical questions where accessibility analysis comes in:

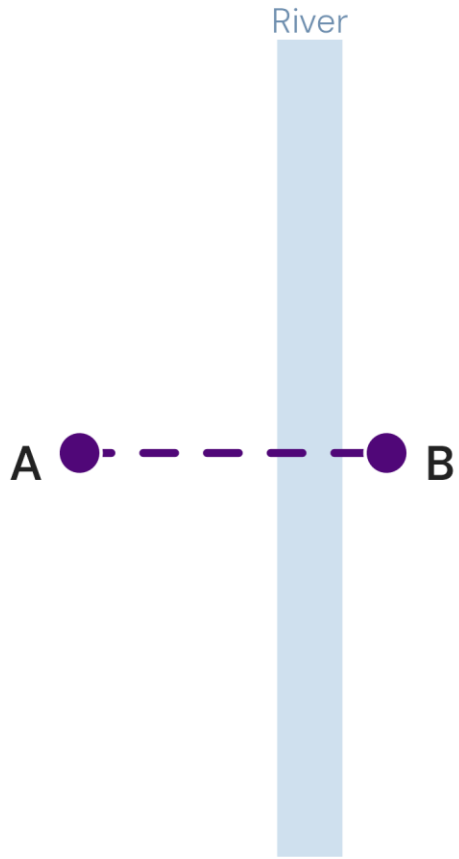
- How many jobs / shops / people can I reach within 15 / 30 / 60 minutes of travel?
- How long do I need to travel to reach N jobs / shops / people?
- How does accessibility differ spatially?
- How does accessibility differ temporally?

What accessibility analysis needs

To quantitatively measure accessibility in a GIS we need at least:

- A set of origins (e.g. homes or schools).
- A set of destinations (e.g. shops, jobs or fast-food outlets).
- Some form of a digital spatial network to connect origins and destinations.

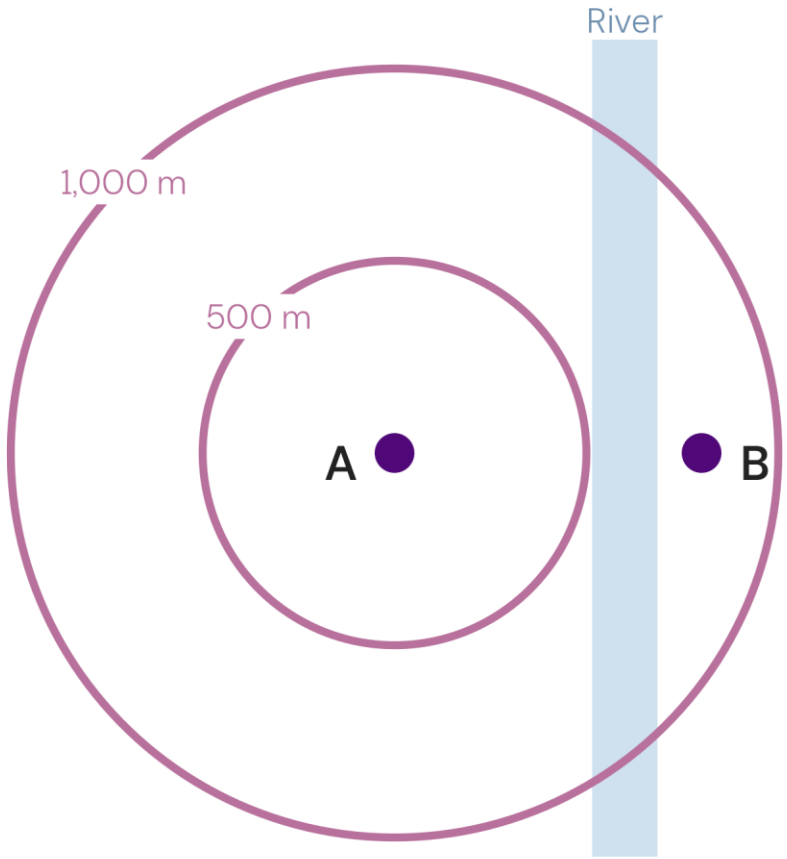
Accessibility of B from A



Straight line from A to B

800 m

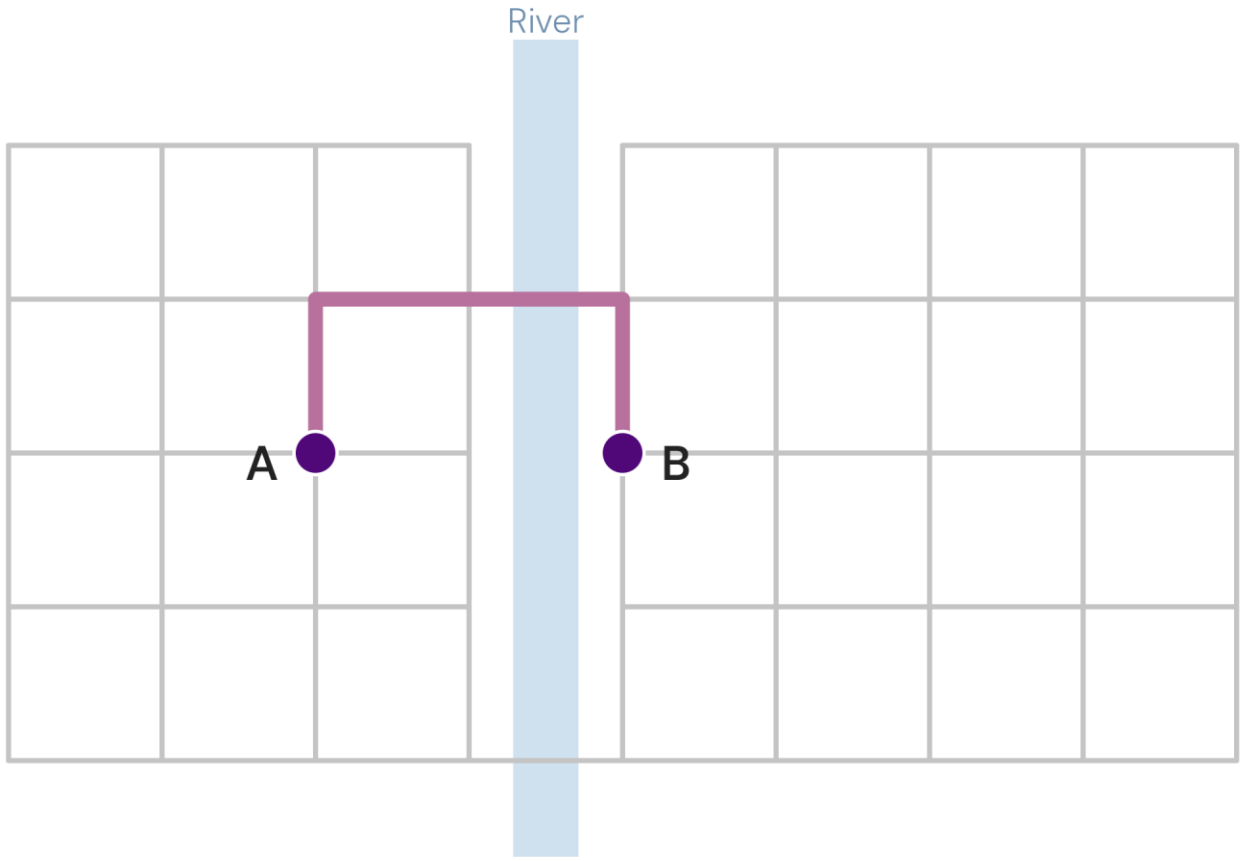
A buffer measures straight-line distance



B lies inside the

1,000 m buffer

The network gives the real distance



Straight line

800 m

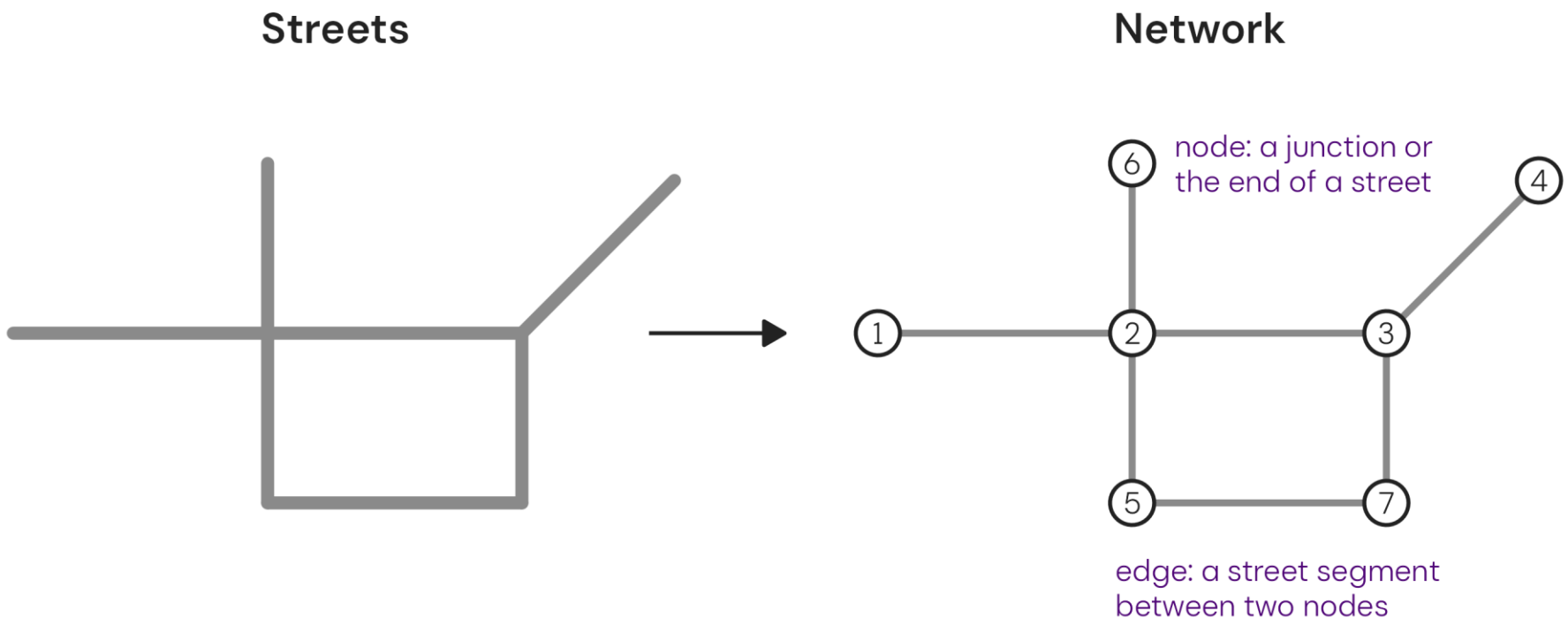
Route along the streets

1,600 m

Spatial network

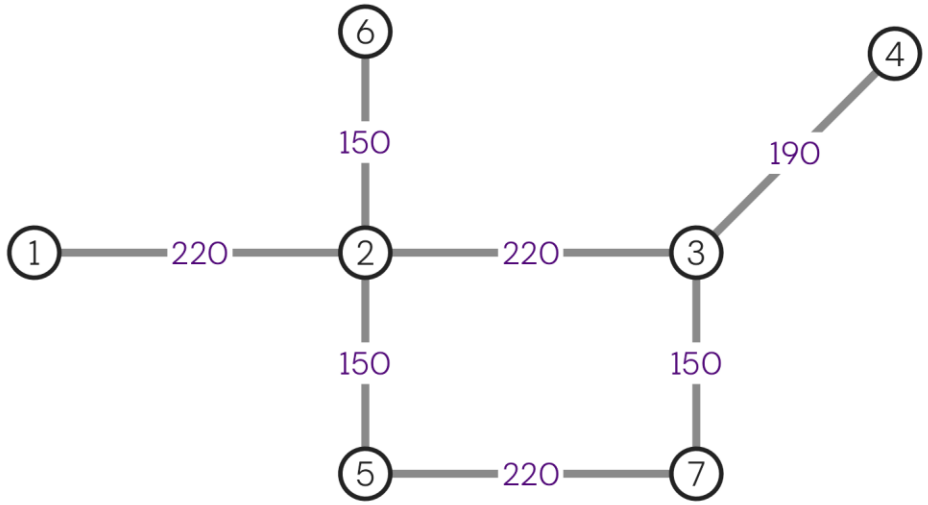
- A spatial network is an organised system or collection of nodes and edges embedded in geographic space.
- Nodes can be a representation of physical objects in geographical space, and edges show what connections are formed between the objects.
- Examples of networks: street configuration, transportation and shipping routes, river basins, telecommunication lines, etc.

A network is nodes and edges



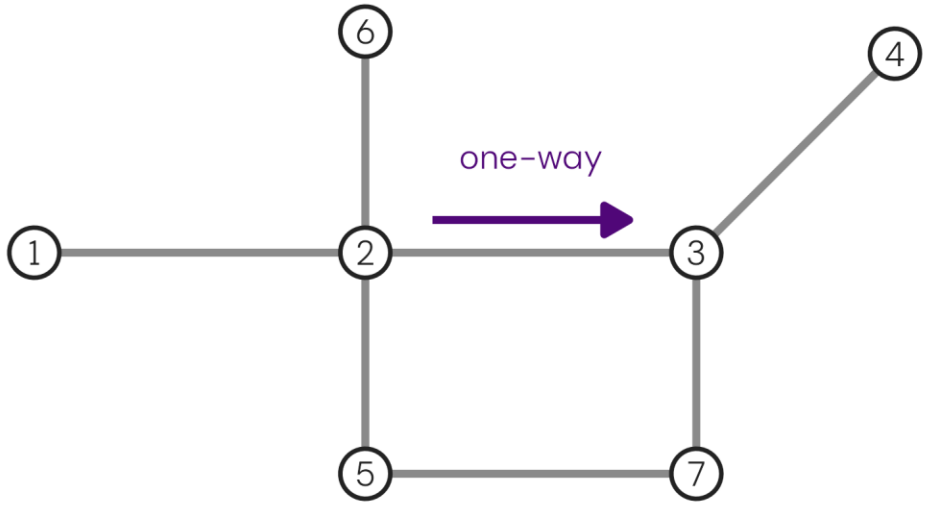
Edges carry a cost

Each edge carries a cost: here its length in metres



Edge	Length (m)	Walk (min)
1 – 2	220	2.9
2 – 3	220	2.9
3 – 4	190	2.5
2 – 6	150	2.0
5 – 7	220	2.9
7 – 3	150	2.0
2 – 5	150	2.0

Edges can have a direction



From 2 to 3

220 m

From 3 back to 2

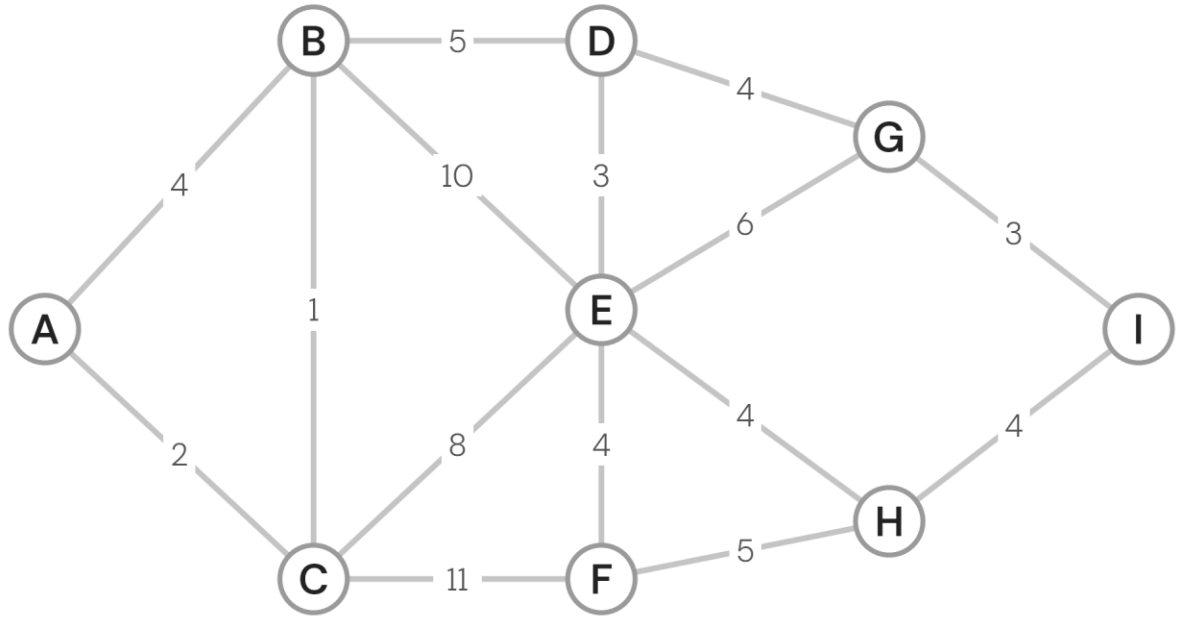
520 m

The way back goes round the block via 7 and 5

Dijkstra’s algorithm: cheapest routes

- Every edge has a cost: distance, travel time or something else.
- Start at the origin with cost 0. Every other node is not yet reached.
- Settle the cheapest node that is not yet settled, then update the costs of its neighbours.
- Stop when the destination is settled, or continue to reach every node.

A network with a cost on every edge

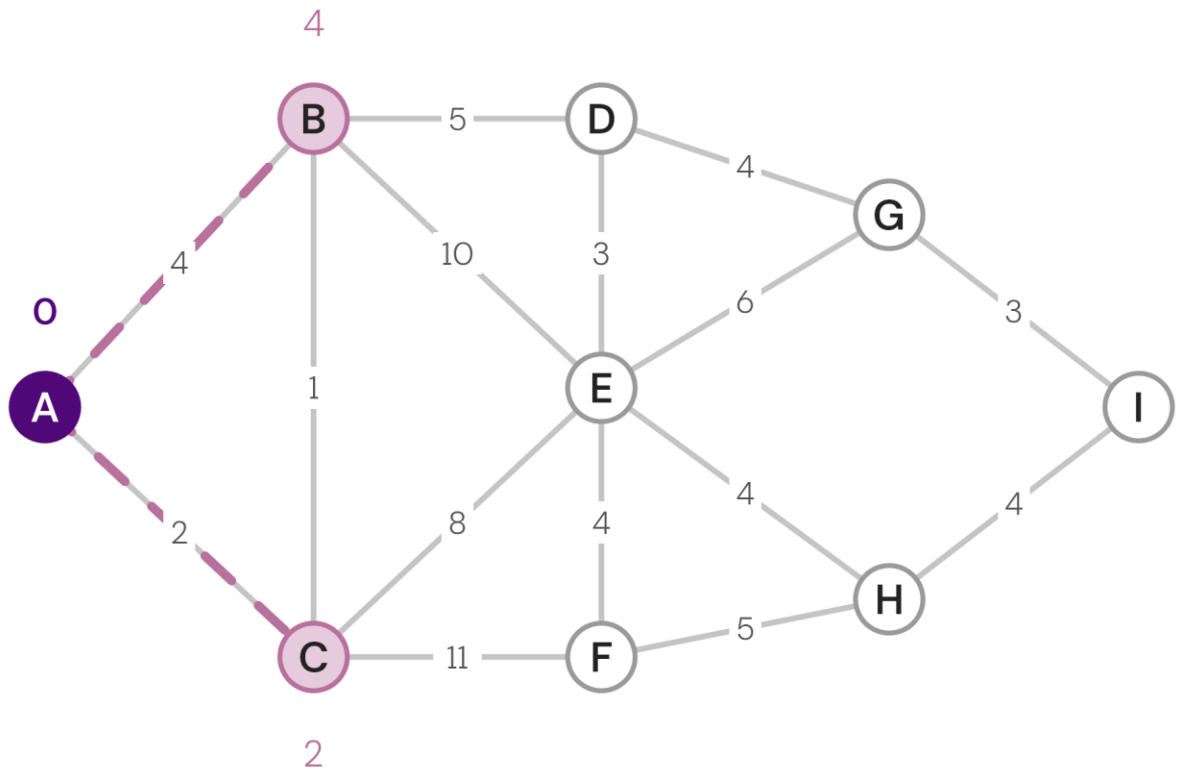


Settled (final cost)

Candidates (best cost so far)

A 0 (the origin)

Step 1: settle the origin, label its neighbours



Settled (final cost)

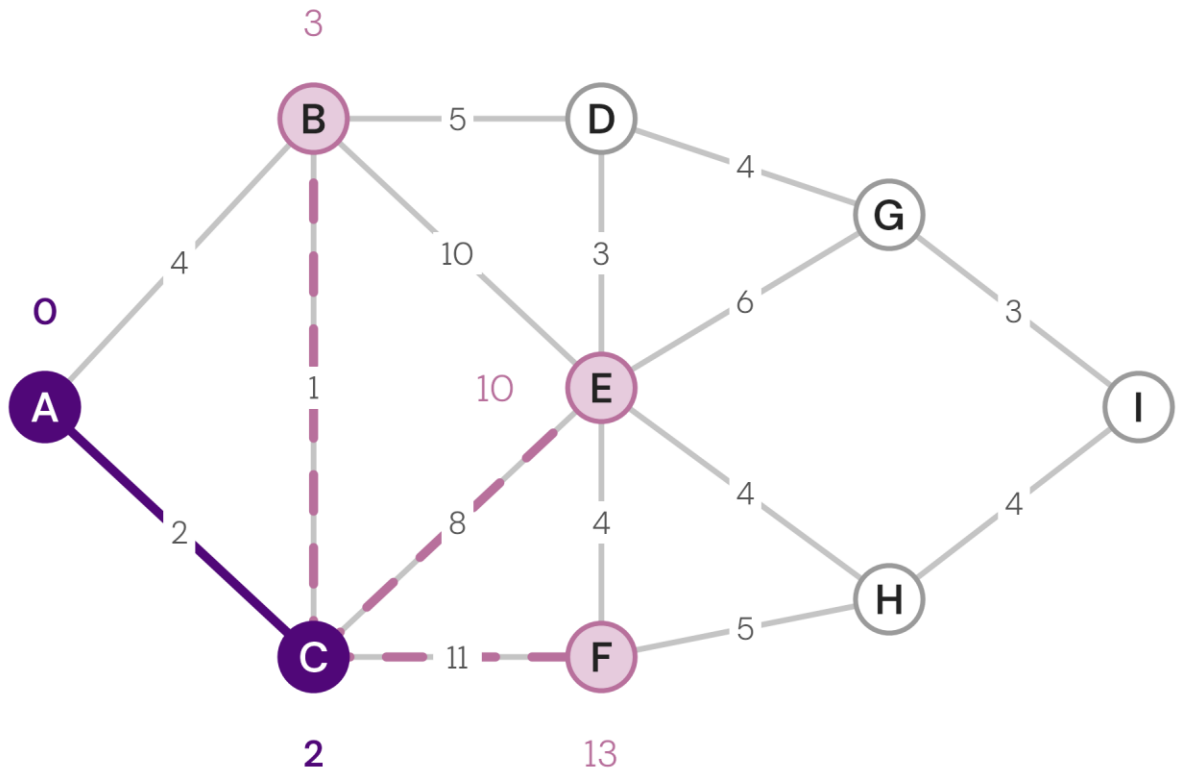
A 0

Candidates (best cost so far)

B 4 (via A)

C 2 (via A)

Step 2: settle the cheapest candidate



Settled (final cost)

A 0

C 2

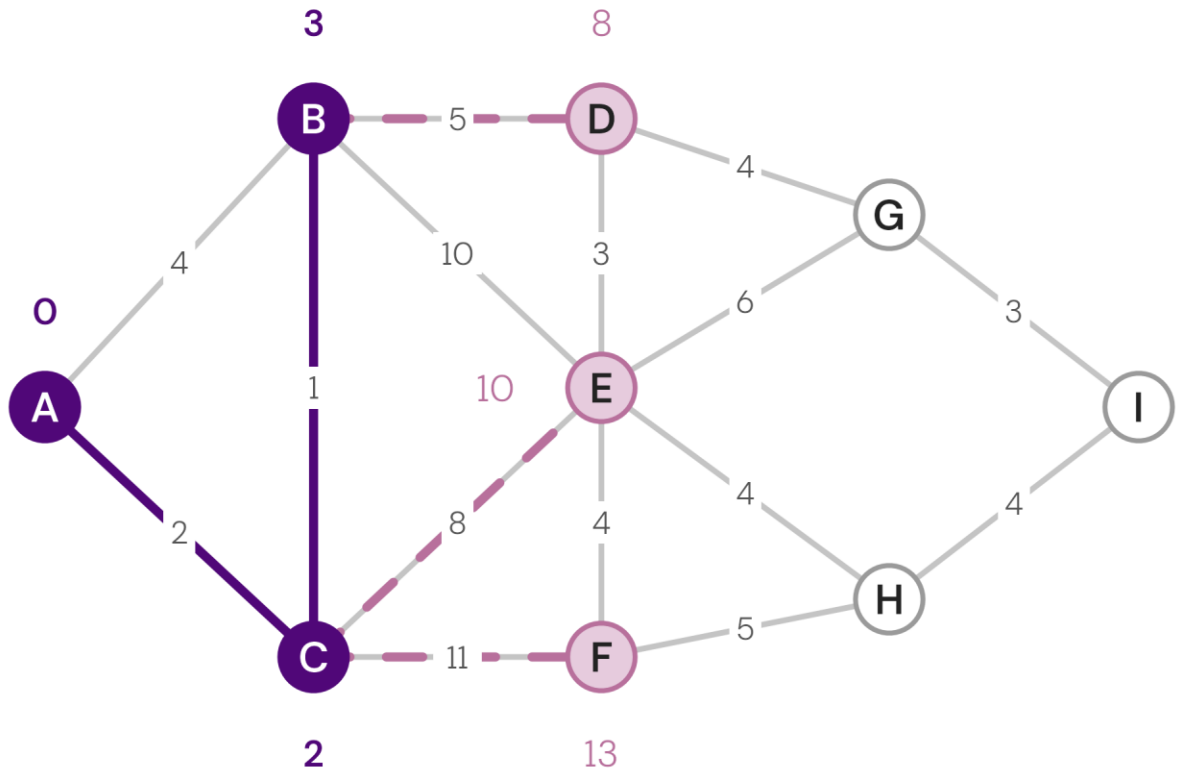
Candidates (best cost so far)

B 3 (via C)

E 10 (via C)

F 13 (via C)

Step 3: repeat with the next cheapest



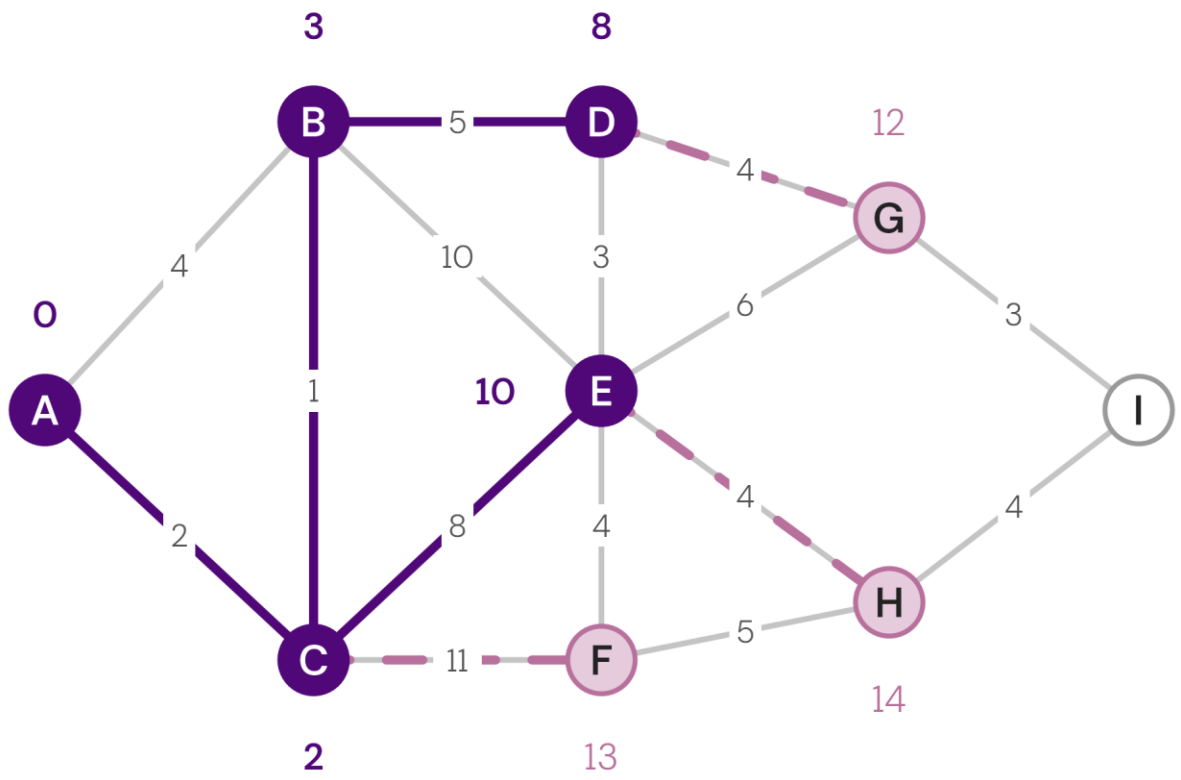
Settled (final cost)

A 0
C 2
B 3

Candidates (best cost so far)

D 8 (via B)
E 10 (via C)
F 13 (via C)

Step 4: keep going



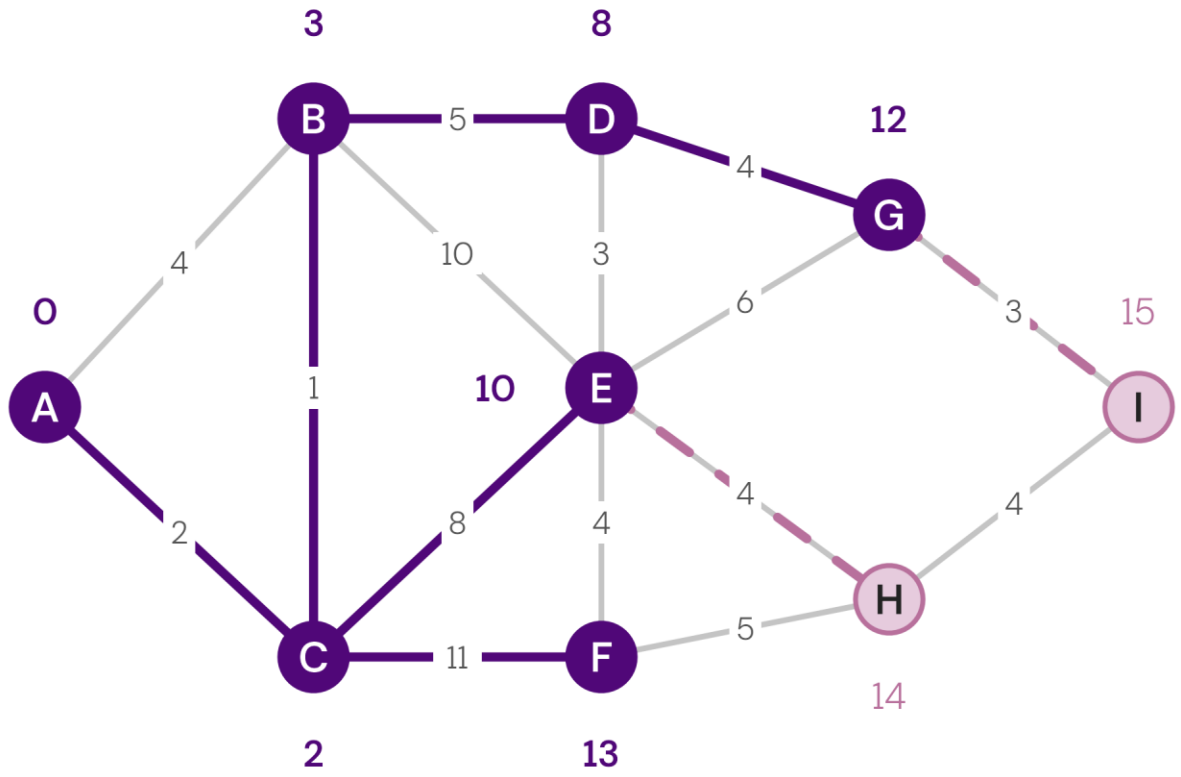
Settled (final cost)

A 0
C 2
B 3
D 8
E 10

Candidates (best cost so far)

F 13 (via C)
G 12 (via D)
H 14 (via E)

A labelled destination is not yet settled



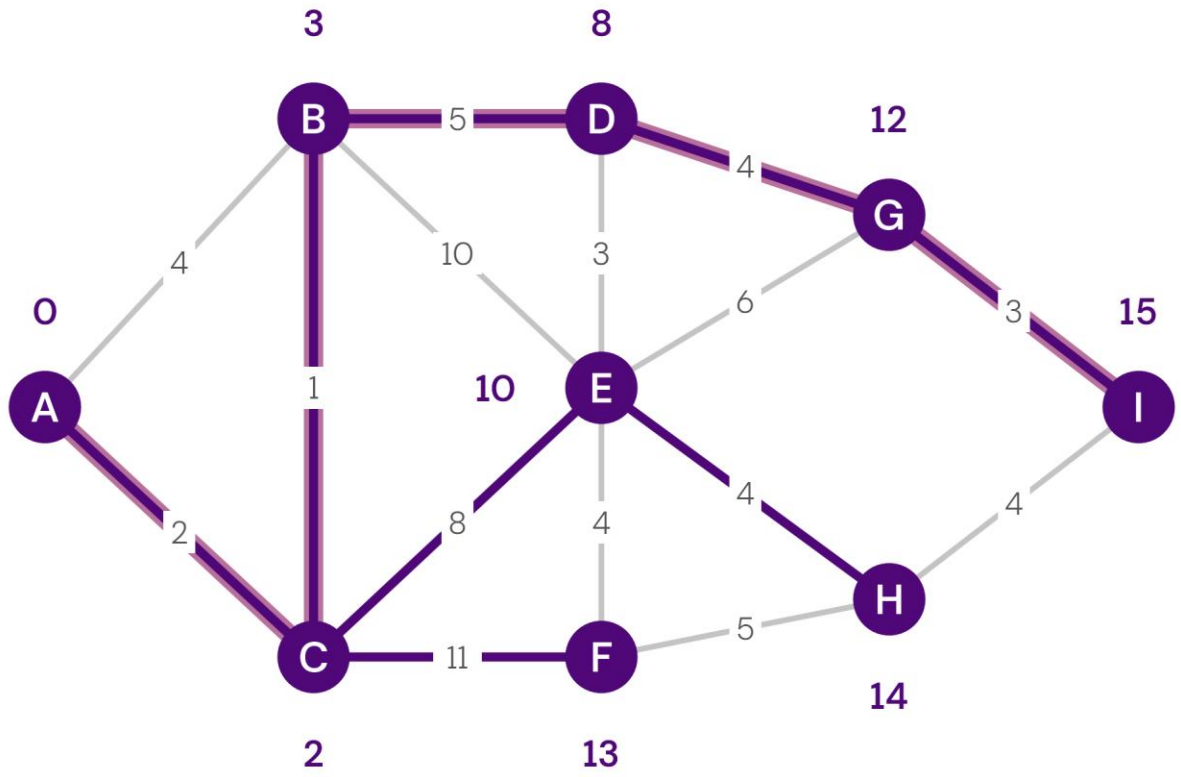
Settled (final cost)

A 0
C 2
B 3
D 8
E 10
G 12
F 13

Candidates (best cost so far)

H 14 (via E)
I 15 (via G)

Result: the cheapest route from A to I costs 15



Settled (final cost)

A 0
C 2
B 3
D 8
E 10
G 12
F 13
H 14
I 15

Candidates (best cost so far)

none left

From distances to an accessibility score

Destinations (fast-food outlets)

Origins (schools)	F1	F2	F3	F4	F5	F6	Within 500 m
S1	1,200	**260**	1,480	**200**	**430**	**140**	**4**
S2	**310**	1,390	**480**	970	510	**380**	**3**
S3	1,080	**260**	1,040	**360**	810	1,190	**2**
S4	**370**	**360**	1,410	**450**	990	1,400	**3**

Network distance in metres. Shaded: within 500 m, counted in the score.

Accessibility with Dijkstra: what we need

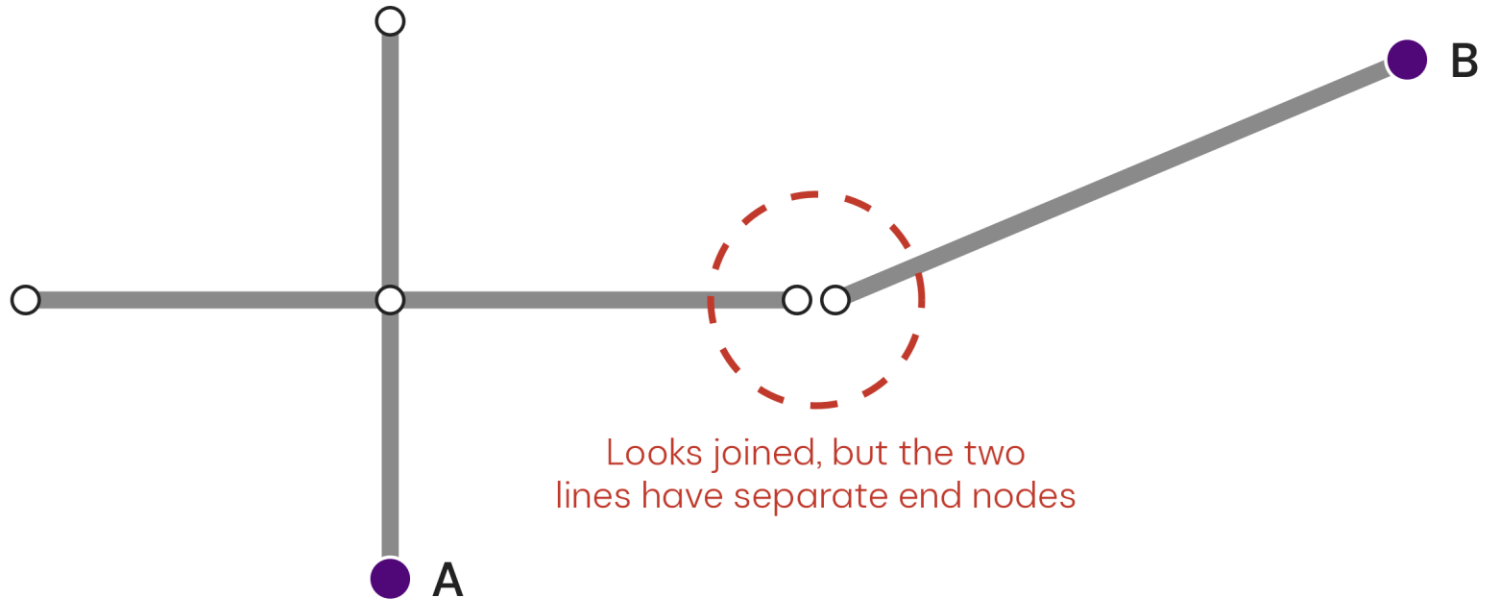
… with Dijkstra’s algorithm we need

- A set of origins (e.g. homes or schools).
- A set of destinations (e.g. shops, jobs or fast-food outlets).
- Some form of a digital spatial network to connect origins and destinations.
- Impedance values per mode of transport / costs for each network segment
- If available: access indicators to construct a weighted graph

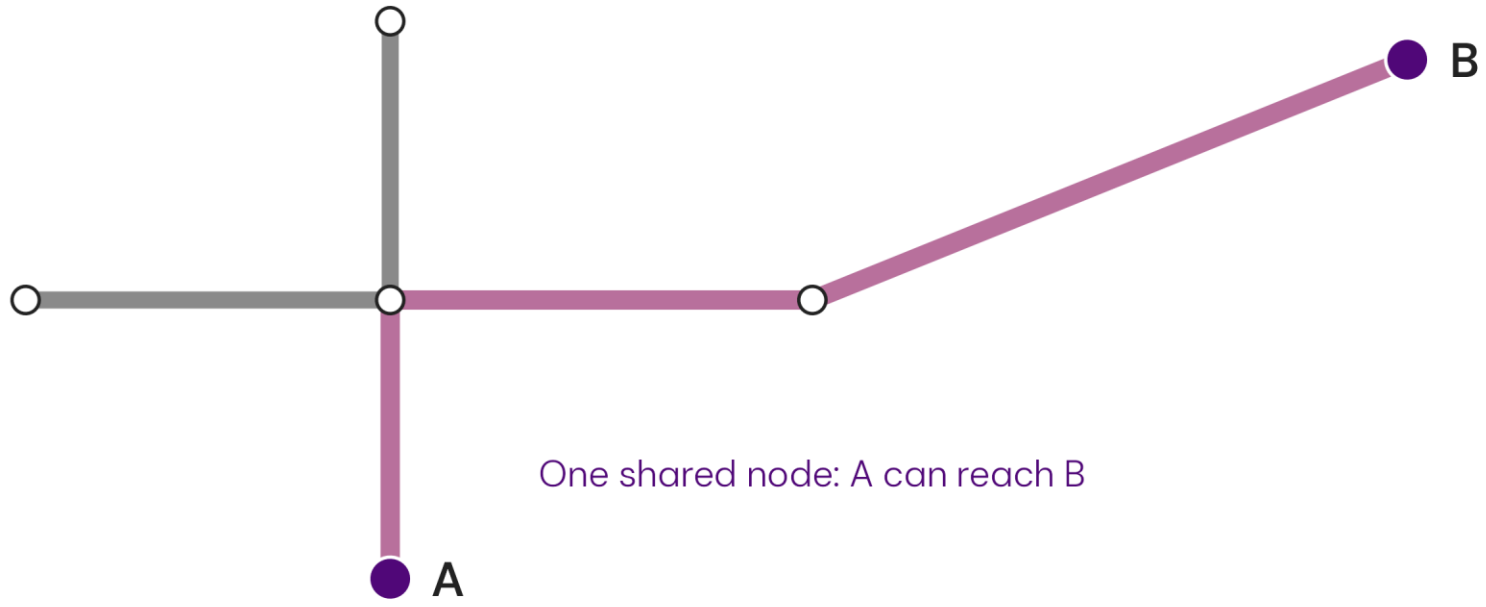
Checking a network: five questions

- Completeness – are all areas covered?
- Attributes – are they correct?
- Connectivity – are all network segments that should be connected, connected?
- Topology – are all network segments connected the way they should be?
- Coverage – is the full network covered?

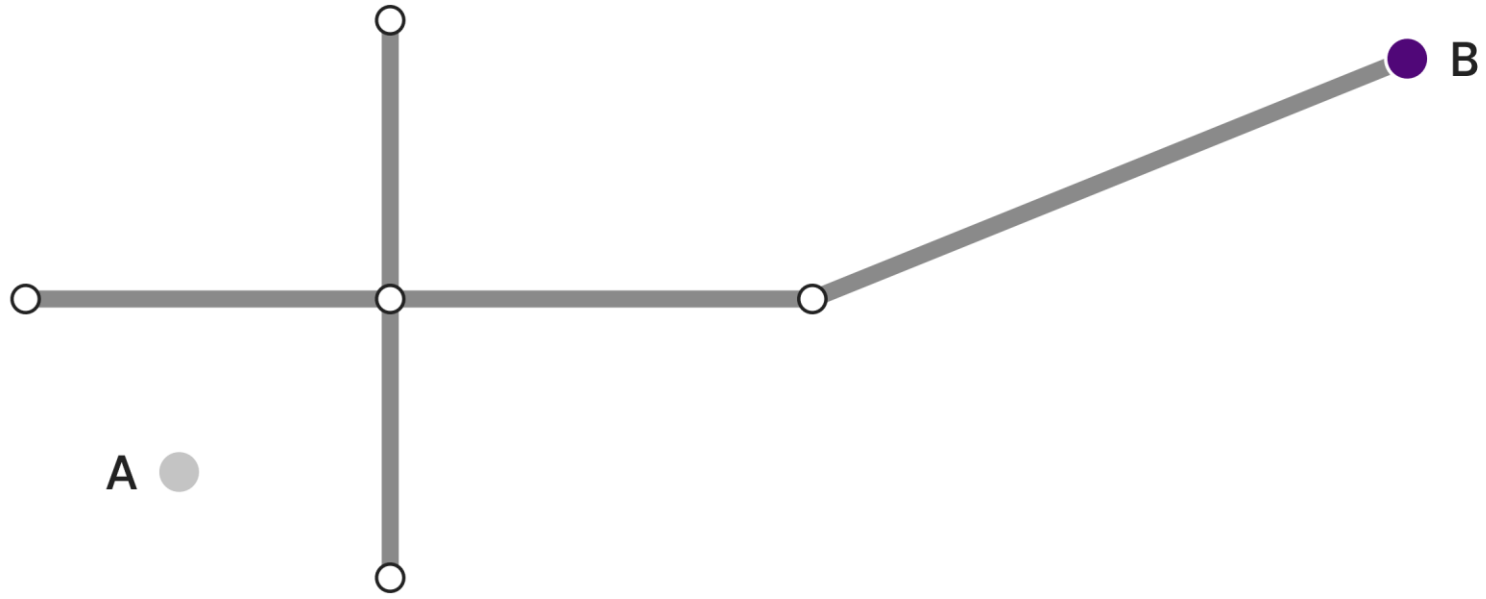
Connectivity: lines that look joined may not be



Fix it with one shared node

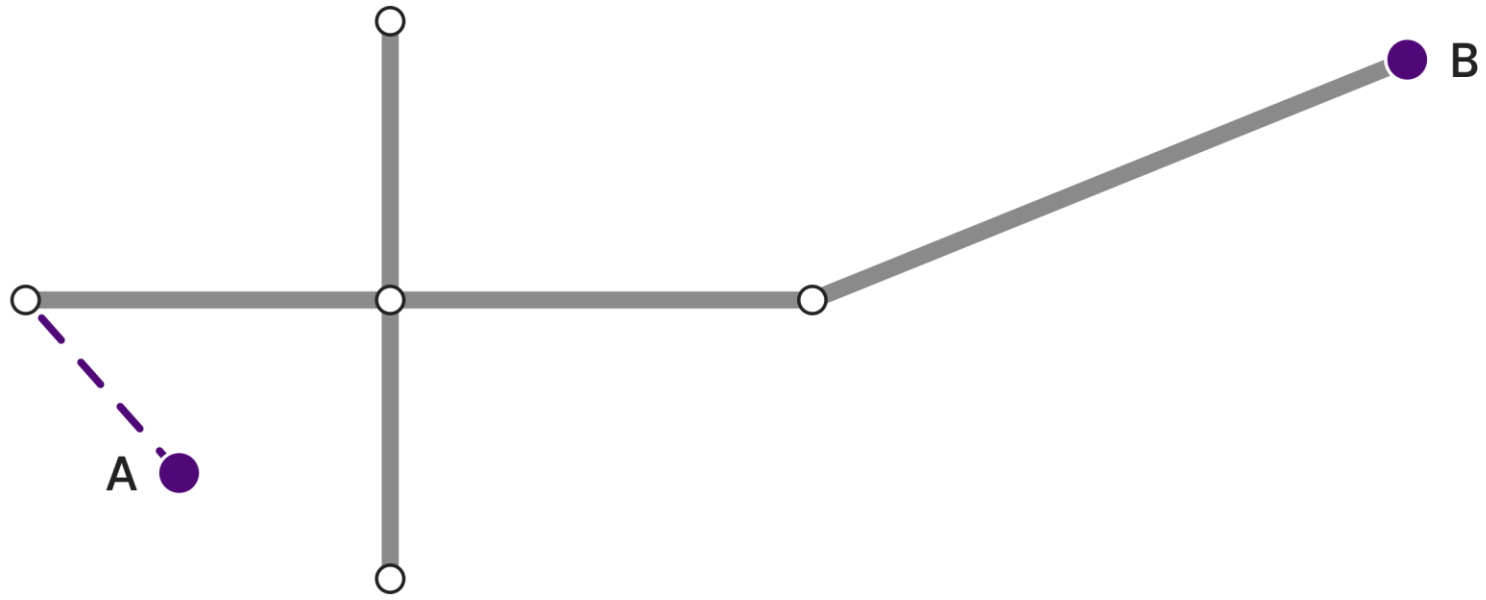


Origins and destinations must be on the network



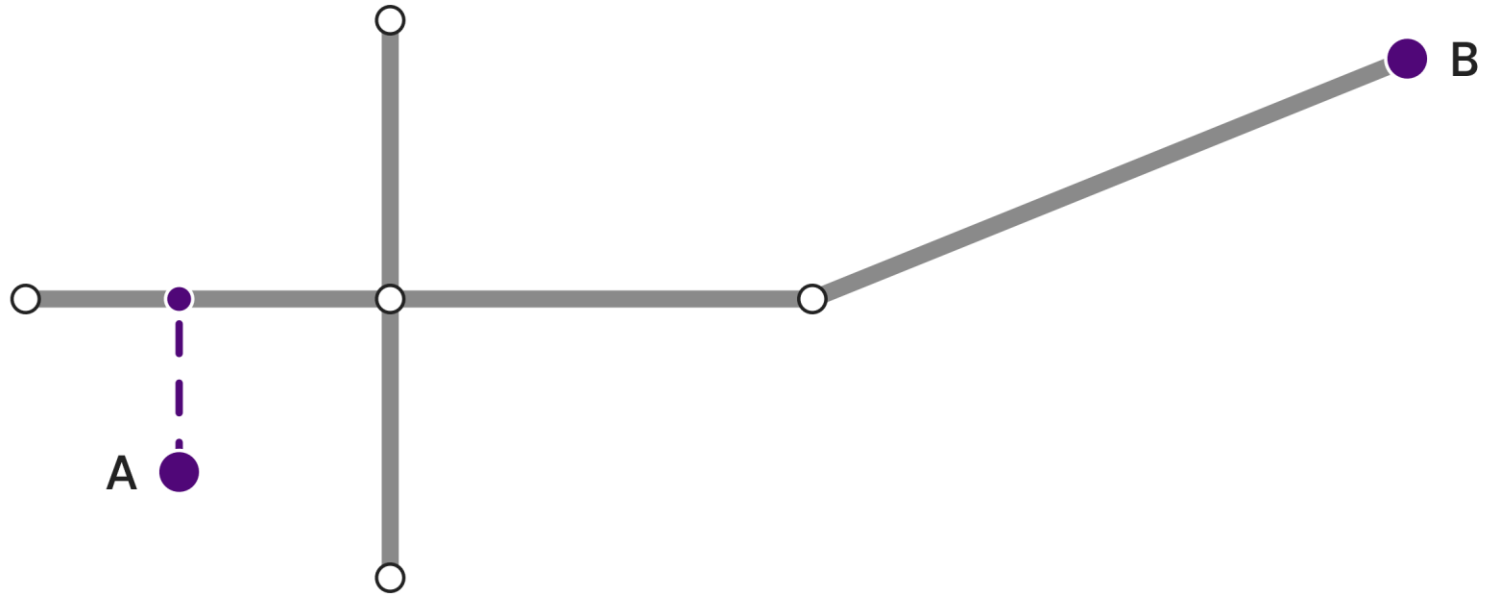
A is not on the network:
there is nothing to route from

Option 1: snap to the nearest node



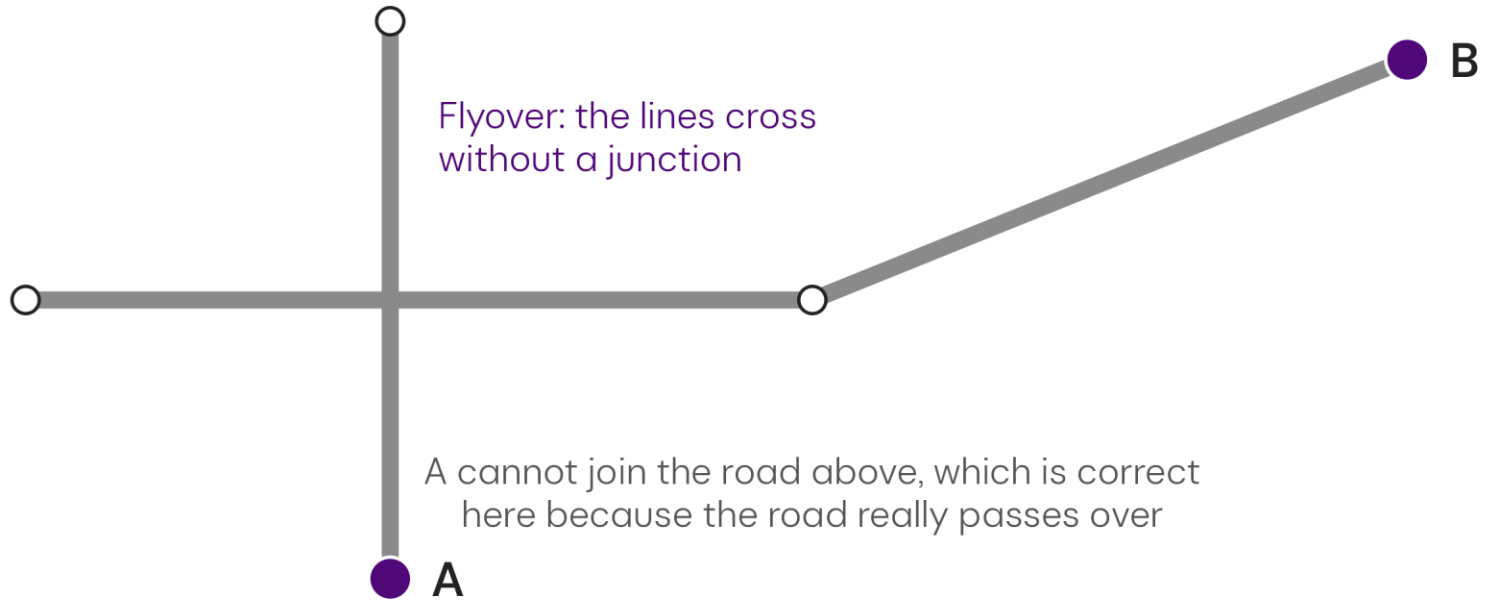
Snap A to the nearest node (120 m away)

Option 2: snap to the street with a feed link

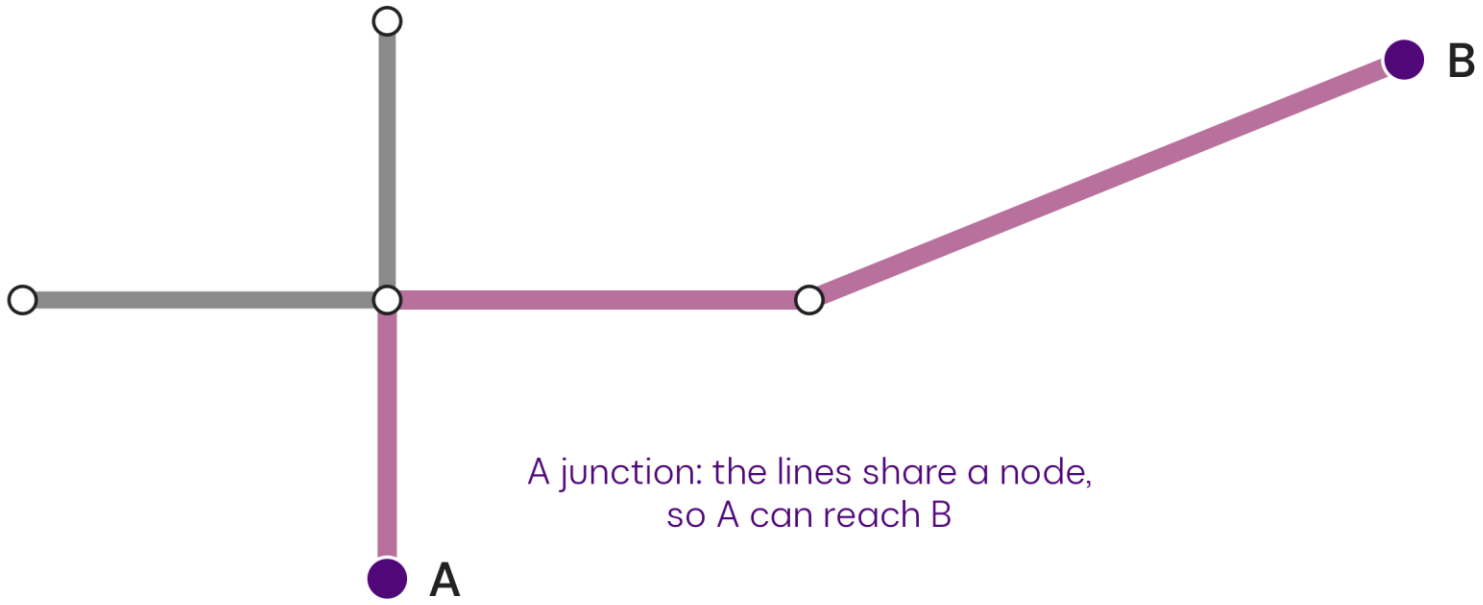


Feed link: a new node on the street, 90 m from A, with its own cost

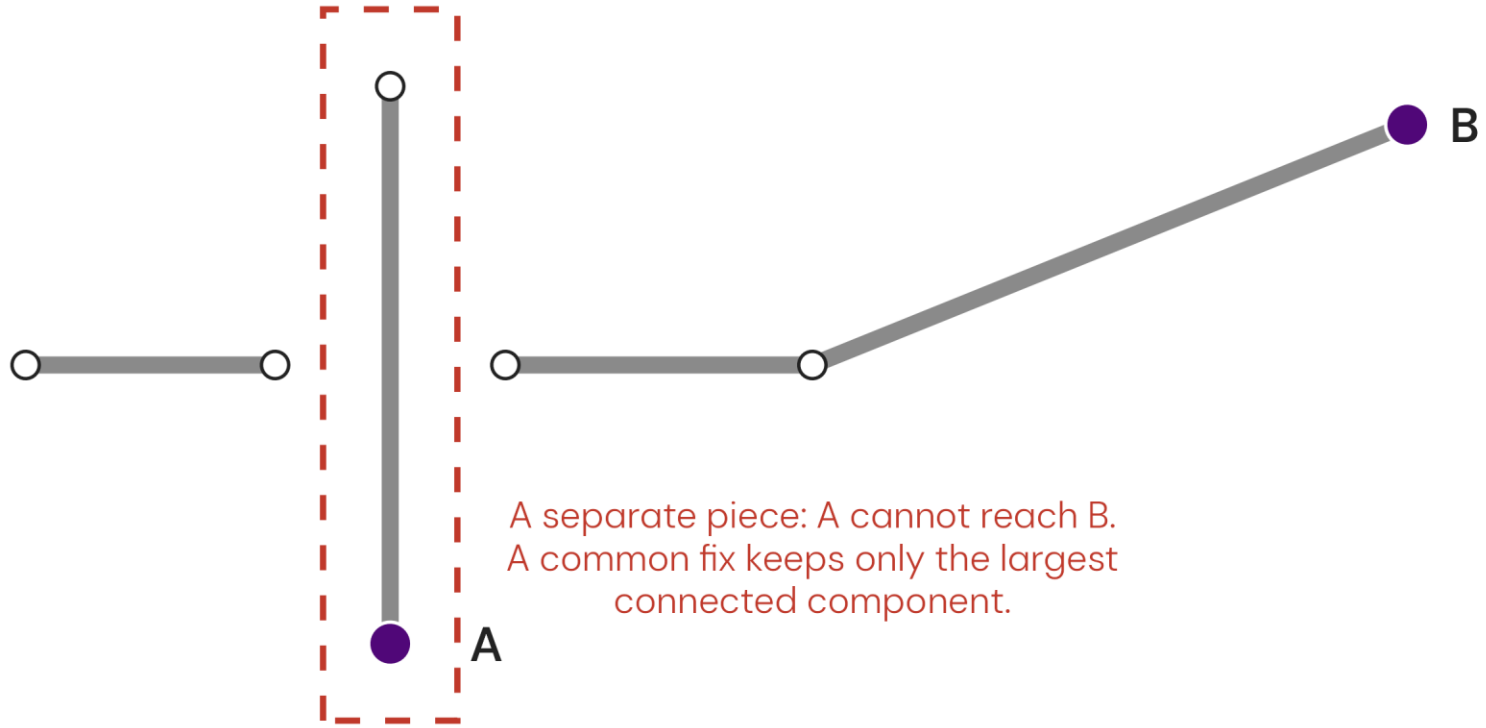
Topology: lines can cross without connecting



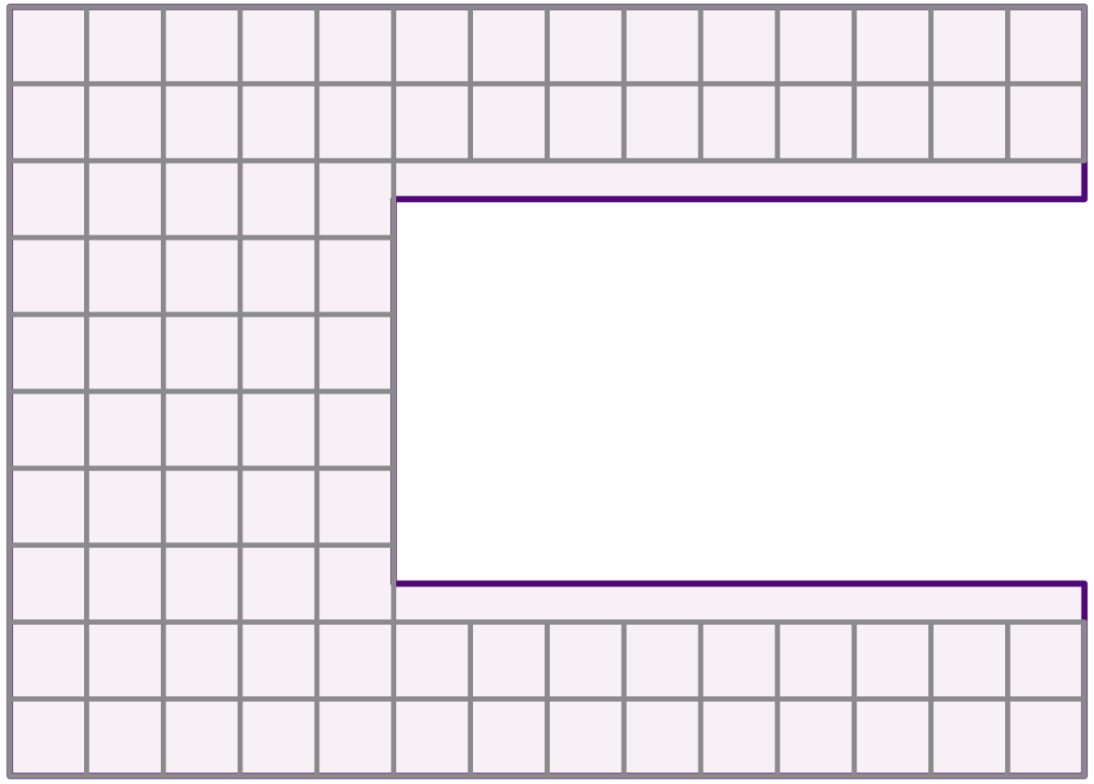
At a junction, the lines must share a node



Disconnected pieces cannot reach each other

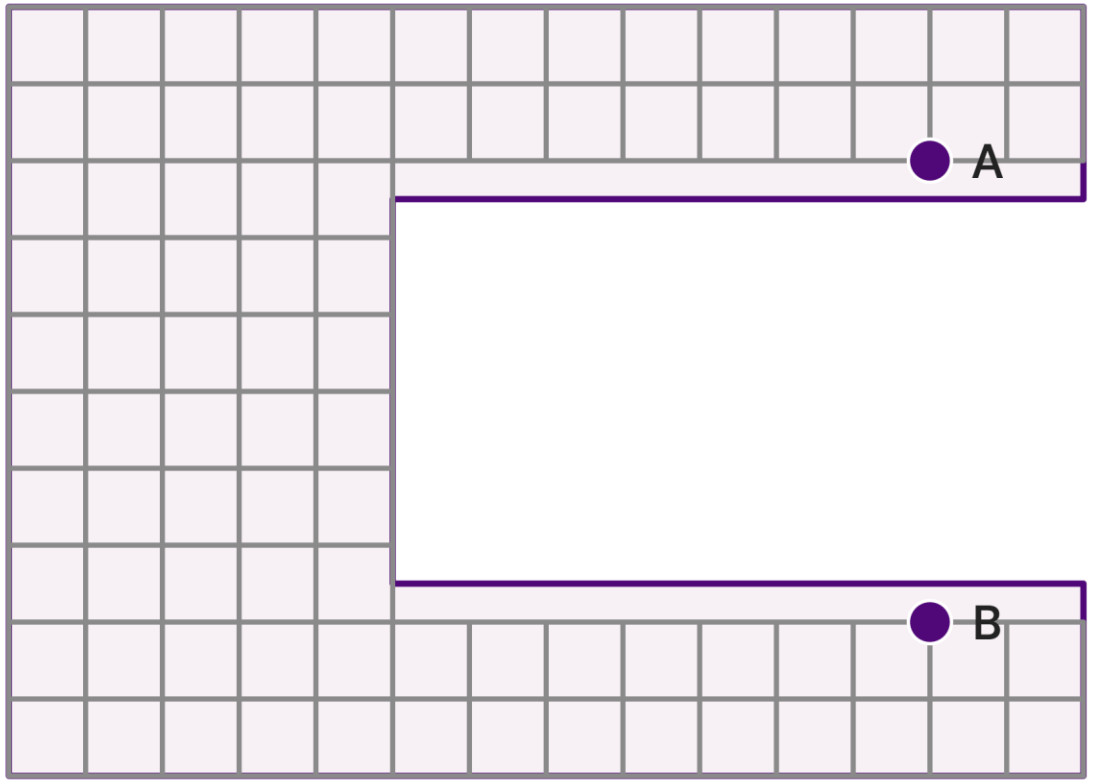


Coverage: does the network cover the study area?



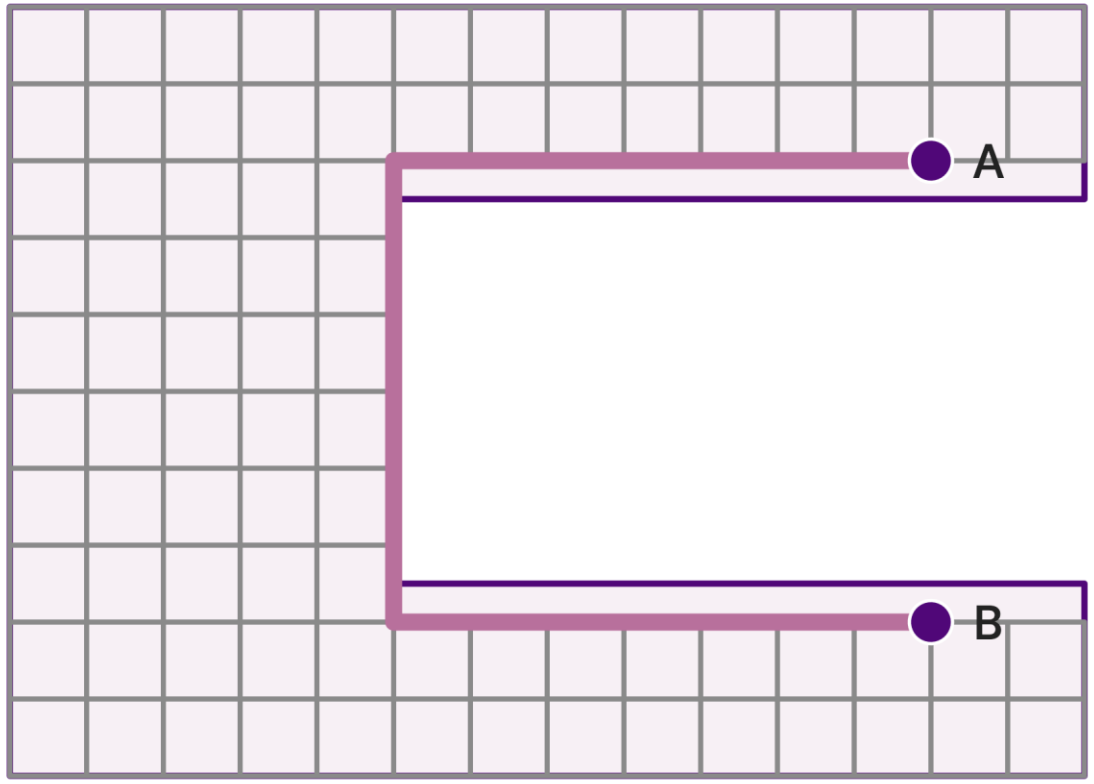
The roads stop at the boundary of the study area

Close locations can be far apart inside the area



A and B are close,
but the study area
wraps around a gap

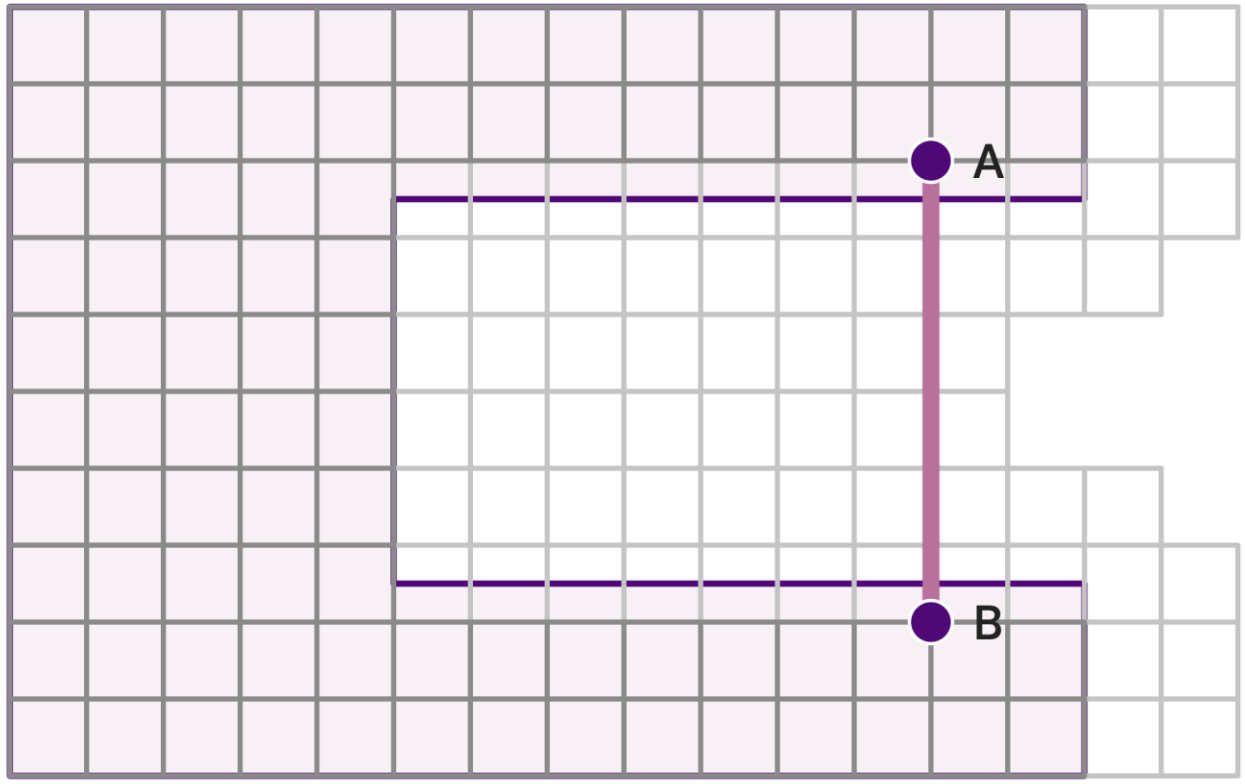
A clipped network forces a long detour



Route from A to B
along the roads inside
the study area

2,000 m

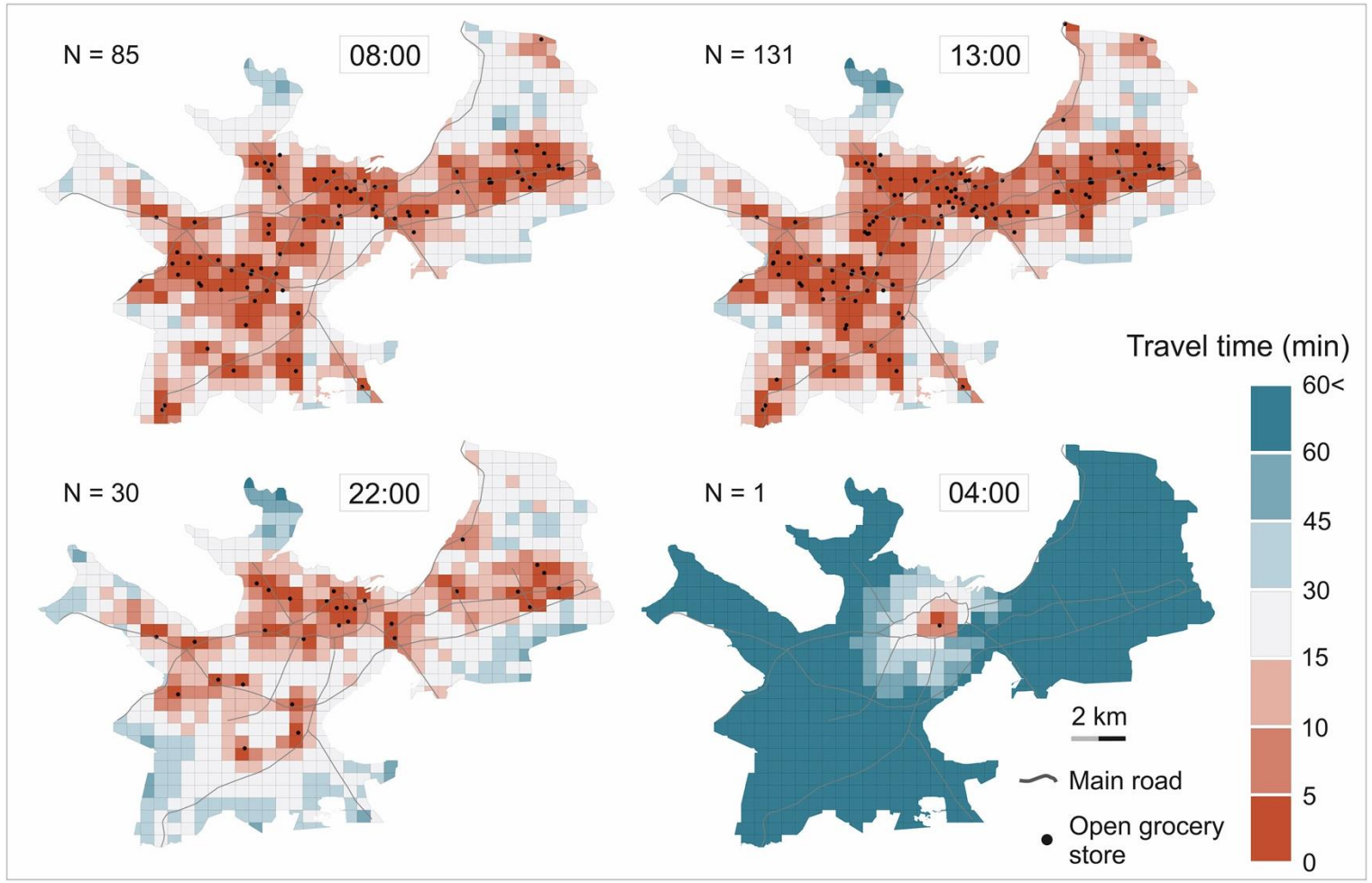
Include roads outside the area



Route when roads up to 250 m outside the area are included

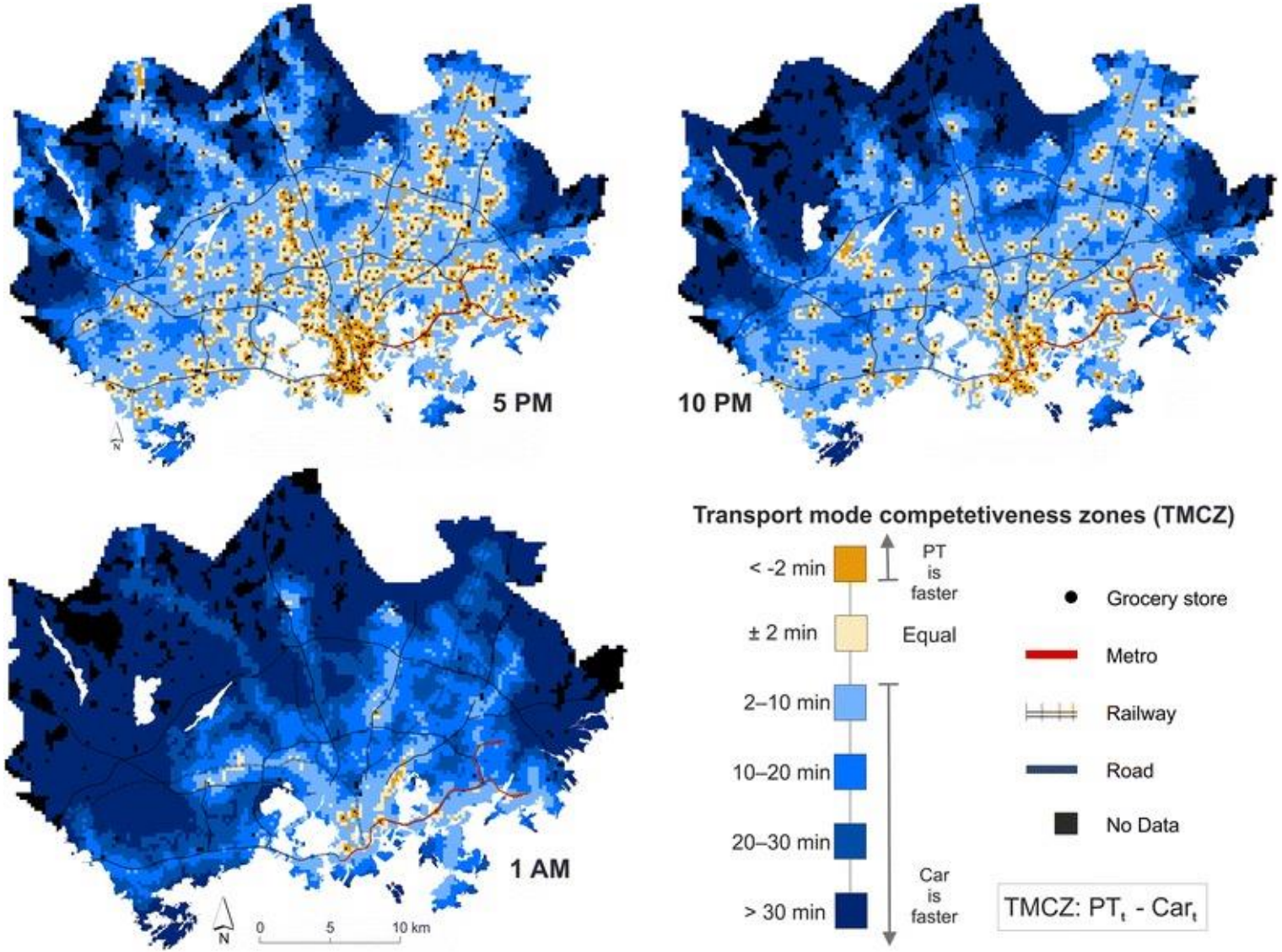
600 m

Accessibility changes through the day



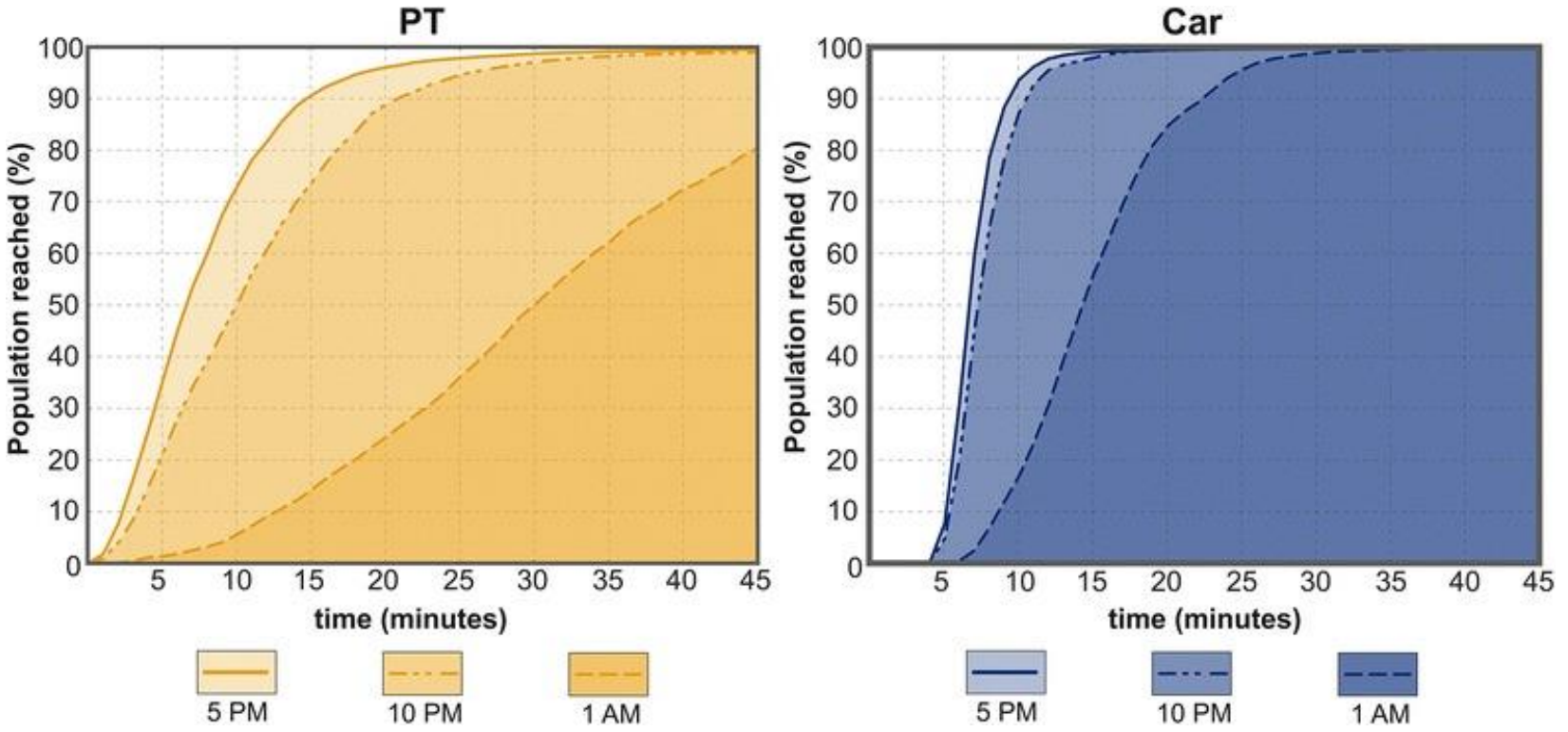
Järv *et al.* 2018

Accessibility depends on the mode of transport



Tenkanen *et al.* 2016

Time of day and mode both matter



Tenkanen *et al.* 2016

Conclusion

- Accessibility needs to be operationalised: accessible to what, for whom and when?
- We need origins, destinations and a network.
- Dijkstra’s algorithm finds the cheapest route from an origin to every node.
- Network quality matters: connectivity, topology and coverage can change the result.
- Open-source tools: dodgr, R5 (r5r, r5py), stplanr.

Questions

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